
STEERING THE HERD: A FRAMEWORK FOR LLM-BASED CONTROL OF SOCIAL LEARNING

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ABSTRACT

Algorithms increasingly mediate what people learn, from social media feeds and targeted advertising to LLM-powered assistants. This paper makes three contributions to understanding that mediation when users also learn from one another. First, we introduce a controlled sequential social-learning model in which an information-mediating planner, such as an LLM, chooses the precision of agents’ private signals while agents update from the observed actions of predecessors. The same framework covers an altruistic planner that seeks to improve social welfare and a biased planner that seeks to induce its preferred action. Second, we characterize the resulting dynamic optimization problem: we prove convexity of the altruistic value function and derive belief-dependent optimal policy regimes for altruistic and biased planners. These regimes make the mechanism visible: depending on public belief, an optimal planner may invest maximally, stop investing, increase or decrease precision with belief, and, for a biased objective, intentionally obfuscate signals. Third, we evaluate the framework with LLM-based simulations in which LLMs act as planner, agents, and oracle. Even under transparency constraints—information parity with individuals, no lying or cherry-picking, and fully observable controls—information mediation substantially shifts social welfare in either direction. The LLM planner exhibits emergent strategic behavior that broadly mirrors the theoretical regimes, while deviations point to non-Bayesian belief updating by LLM agents. Together, these results provide a tractable and empirically grounded framework for studying the impact and regulation of LLM information mediators.

1 INTRODUCTION

Large Language Models (LLMs) are becoming consequential information mediators in settings where individual choices aggregate into social outcomes: search engines and news aggregators (Zhang et al., 2024a; Gao et al., 2024), personal assistants that offer medical or political advice (Huo et al., 2025; Haupt & Marks, 2023; Schiele et al., 2024; Argyle et al., 2023), and recommender systems that shape what users see next. At the same time, LLMs have demonstrated persuasive (Potter et al., 2024; Carrasco-Farre, 2024; Rogiers et al., 2024), manipulative (Williams et al., 2025; Liu et al., 2025; Jones & Bergen, 2024), and strategic (Lorè & Heydari, 2024; Payne & Alloui-Cros, 2025; Zhang et al., 2024b) behavior in user interactions. The contribution of this paper is to study such influence not as a collection of isolated one-on-one interactions, but as a dynamic process in which algorithmic mediation interacts with social learning among users.

A simple recommendation example illustrates the mechanism we formalize. Suppose an LLM-powered system recommends a new restaurant to a sequence of potential customers. The restaurant’s quality—the state of the world—is either good or bad, and neither the users nor the system observes that state directly. Each user receives a personalized recommendation, interpreted in our model as a private signal, and then decides whether to patronize the restaurant. The recommendation system, which we call the *planner*, can invest resources to make that signal more informative: for instance, the LLM may generate a highly tailored review using the user’s known preferences. This captures realistic costs of mediation, such as market research, surveys, or data acquisition required to tailor information effectively.

The distinctive feature of our setting is that each recommendation also affects future users through public behavior. A user does not rely only on the current private signal; she also observes earlier users’ actions. If the previous ten users patronized the restaurant, the next user reasonably infers that they likely received favorable information. This *social learning* channel shapes decisions in domains such as vaccine adoption (e.g., Rao et al. (2007); Bauch & Bhattacharyya (2012)), new technology adoption (e.g., Gillingham & Bollinger (2021); Weber (2012)), and political or moral opinion formation (e.g., Brady et al. (2021); Guilbeault et al. (2018)). Our first contribution is to place an information-mediating planner inside this sequential social-learning process, where each precision choice affects the current agent and creates an information *externality* through the public belief faced by future agents.

We make this contribution through a tractable model of *controlled social learning*. In the model, the planner does not lie, censor, or cherry-pick information; it only chooses, at a cost, how precise each agent’s private signal will be. This restriction is deliberate because it isolates a subtle but important form of algorithmic influence: steering collective beliefs by changing informativeness rather than content. The model lets us ask reviewer-facing questions directly: How should a planner use signal precision to steer public belief? When does social learning amplify or dampen that control? What welfare consequences follow from the planner’s objective?

The planner’s objective is central to our contribution structure. An *altruistic* planner, such as an ideal recommendation system, wants agents to take the correct action: patronize the restaurant if and only if it is good. A *biased* planner, such as a recommender receiving a kickback from the restaurant, wants to induce one action—patronizing—regardless of the state. By analyzing both objectives in the same model, we separate the mechanics of information control from the normative question of whether that control helps users.

This distinction is related to, but not identical with, alignment. The altruistic objective is aligned with agents’ *expected utility* under uncertainty. The biased objective is state-independent and therefore not generally aligned, although it can coincide with agents’ *realized utility* when the true state favors the planner’s preferred action. We therefore study both aligned and misaligned instances of information mediation, preserving the same social-learning mechanism while varying the planner’s objective.

Contributions and Outline This paper’s contributions are organized as follows:

1. **A formal framework for controlled social learning.** We introduce a dynamic model in which a centralized information planner chooses the precision of private signals while agents retain decentralized control of their actions and learn from predecessors. The framework covers both altruistic and biased planners, making objective alignment an explicit modeling dimension.
2. **Optimal policy characterizations.** We formulate the planner’s problem as a discounted Markov decision problem over the public belief. For the altruistic planner, we prove convexity of the value function and characterize when investment is or is not optimal. For the biased planner, we characterize belief regimes in which the planner invests, reduces precision, or induces cascades, exposing the strategic tradeoffs created by information externalities.
3. **LLM-based empirical evaluation.** We simulate the framework with LLMs serving as agents, planner, and oracle. The experiments show that accounting for social learning materially changes planner utility and social welfare; that LLM planner policies largely track the analytical regimes; and that deviations are informative about non-Bayesian LLM agent behavior.

We review related work in Section 2, present the model in Section 3, and derive optimal policies in Sections 4 and 5. Section 6 tests the framework with LLM-based simulations, and Section 7 summarizes the implications, limitations, and future directions. Proofs and additional experimental details appear in the appendices.

2 RELATED WORK

Our contributions connect four literatures: social learning, information design, online persuasion and reinforcement learning, and LLM-mediated economic behavior. We emphasize how our framework differs from each line: it studies a planner with no private informational advantage, agents who keep control of their actions, and sequential dependence generated by social learning rather than by hidden environmental parameters.

Social Learning The social-learning literature supplies the sequential belief dynamics that our planner controls. In the canonical models of Bikhchandani et al. (1992) and Banerjee (1992), agents estimate an unknown state from private information and predecessors’ actions; later work characterizes these dynamics in broader environments (e.g., Smith & Sørensen (2000); Arieli & Mueller-Frank (2021)) and studies variations in observed social information, biases, and misperceptions (see Dasaratha & He (2022); Bistriz et al. (2022); Eyster & Rabin (2010) as well as Bikhchandani et al. (2024) and the references therein). Our contribution is not a new uncontrolled learning dynamic, but a control problem layered on top of this dynamic.

Work on the *control* of social learning is comparatively limited (Wei & Anastasopoulos, 2022; Smith et al., 2021; Krishnamurthy, 2012; Bhatt & Krishnamurthy, 2021). The closest papers are Wei & Anastasopoulos (2022) and Smith et al. (2021): Wei & Anastasopoulos (2022) uses two-way communication in which agents report private signals to a planner in exchange for recommendations, while Smith et al. (2021) studies a benevolent planner who directly alters agents’ choice rules and derives a decentralized pivot mechanism. Our model contributes a different form of control for information-mediating algorithms: the planner changes signal precision, receives no direct reports of private signals, and leaves agents’ actions decentralized.

Information Design Our planner’s choice of signal precision is a constrained information-design problem (Bergemann & Morris, 2019). Classical Bayesian persuasion, beginning with Kamenica & Gentzkow (2011), studies a sender who chooses the signal structure observed by a rational Bayesian receiver. In contrast, our planner repeatedly chooses a costly binary symmetric signal for each agent, and the value of that choice depends on how the agent’s action updates the public belief for future agents. The main contribution relative to standard information design is therefore the integration of dynamic social-learning externalities.

Only a small set of papers studies information design in sequential social-learning settings (Arieli et al., 2022; Wu et al., 2025). These works choose an information structure once at the beginning of the sequence and then hold it fixed. Arieli et al. (2022) studies a self-interested sender with a general Bayesian-persuasion signal structure, and Wu et al. (2025) restricts the structure to be Blackwell more informative while maximizing the probability of a correct cascade. Our setting contributes a dynamic counterpart: the sender chooses a new signal precision for each agent, generating richer belief-dependent strategies.

Online Persuasion and RL Recent work at the interface of online learning and Bayesian persuasion studies senders who repeatedly persuade short-lived receivers while learning unknown priors or utility functions (e.g. Castiglioni et al. (2020); Bacchiocchi et al. (2024); Castiglioni et al. (2021); Agrawal et al. (2023)). Related dynamic-persuasion and MDP models introduce state transitions, farsighted receivers, history-dependent or promise-based signaling, and reinforcement-learning connections (e.g. Wu et al. (2022); Su et al. (2022); Bernasconi et al. (2023)). Their central contributions are learning guarantees, regret bounds, and approximation algorithms under uncertainty or computational constraints.

Our contribution is complementary rather than an online-learning result. The planner in our model does not learn unknown environmental parameters, does not observe the state or private signals, and does not change the payoff-relevant state. Sequential dependence instead comes from information externalities: each observed action propagates through the public belief that future agents use. This distinction lets us characterize optimal planner policies directly, rather than measuring regret against a hindsight benchmark.

LLMs A growing literature studies LLMs in information-design and persuasion settings (Harris et al., 2025; Cheng & You, 2025; Raifer et al., 2022; Bai et al., 2024; Duetting et al., 2025). These papers primarily analyze one-on-one interactions, so they do not capture the social-learning channel that is central to our framework. The closest methodological connection is Duetting et al. (2025),

who study information design in which a sender chooses both a signal structure and a framing that can affect a possibly non-Bayesian receiver’s posterior belief, using an LLM oracle to optimize over framings. Building on the idea that LLMs can mimic human economic agents (Dillion et al., 2023; Horton, 2023), we adapt parts of this oracle methodology in Section 6. Our empirical contribution is to use LLMs to instantiate a repeated social-learning environment with an LLM planner, LLM agents, and an oracle for controlled signal precision.

3 SOCIAL LEARNING MODEL

The model’s first contribution is to reduce algorithmic information mediation to a controlled social-learning state variable. We consider a planner and a countable sequence of Bayes-rational agents indexed by $i \in \mathbb{N}_{>0}$. At time $i = 0$, nature draws a fixed but *unknown* exogenous state $\omega \in \Omega := \{G, B\}$, with commonly known prior $\mathbb{P}(\omega = G) = b_1$. In the recommendation example from Section 1, G denotes the restaurant being good.

At each time i , the planner chooses how much to invest in the informativeness of agent i ’s private signal. The resulting signal $s_i \in \Omega$ is binary and matches the state ω with probability $q_i \in [0.5, 1]$, i.e., $\mathbb{P}(s_i = \omega) = q_i$. We refer to q_i as agent i ’s signal precision, the control variable that connects the planner’s intervention to the agents’ learning problem.

Conditional on ω , each agent’s signal is independent of other agents’ signals and of the public history. Agent i then combines this private signal with the observed actions of earlier agents $j < i$ and chooses $a_i \in \Omega$. She receives utility 0 when $a_i = \omega$ and utility $-C$ otherwise, with $C > 0$, so every agent maximizes expected utility by trying to match the state. In the running example, this is the decision to patronize the business if and only if the state is G .

3.1 AGENTS’ DECISION PROBLEMS

The key state variable in the framework is the public belief generated by observable behavior. Each agent observes all predecessor actions and their corresponding signal precisions, summarized by $\mathcal{H}_i := (b_1, (q_j, a_j)_{j < i})$. Because the planner and agents share this same information, they also share a *public* belief $b_i = \mathbb{P}(\omega = G | \mathcal{H}_i)$ just before agent i acts; before any action, this belief is the prior b_1 . As in Banerjee (1992) and Bikhchandani et al. (1992), $(b_i)_{i \in \mathbb{N}}$ is a Markov process and a sufficient statistic for the full history $\mathcal{H}_i$ (see Appendix B.3 for proof).

Given $\mathcal{H}_i$ (equivalently, by the Markov property, b_i) and q_i , agent i chooses $a_i \in \Omega$ to maximize expected utility, breaking ties in favor of the action matching s_i . Her *private* belief $\tilde{b}_i$ combines the public belief with her private signal of precision q_i (derivation in Appendix B.1):

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = \begin{cases} \frac{q_i}{1+2b_iq_i-b_iq_i} b_i & s_i = G \\ \frac{1-q_i}{b_i+q_i-2b_iq_i} b_i & s_i = B \end{cases} \quad (1)$$

This posterior yields a simple decision rule: choose G if $\tilde{b}_i > 0.5$, choose B if $\tilde{b}_i < 0.5$, and follow the private signal if $\tilde{b}_i = 0.5$. Substituting for $\tilde{b}_i$ makes the social-learning threshold explicit:

$$a_i = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & q_i < 1 - b_i \end{cases} \quad (2)$$

Agent i ’s action is then the only new public information for later agents. The induced public-belief update is (see Equation (14), Appendix B.2):

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (3)$$

Remark 1. The update rule identifies the channel through which control matters. When $1 - q_i \leq b_i \leq q_i$, Equation (2) implies that agent i ’s action perfectly reveals her private signal, so the next public belief equals agent i ’s private belief. Outside this interval, the public belief is strong enough that the private signal does not affect the action; the action is uninformative, b_i is unchanged, and the process reaches an absorbing information cascade or herding state in the sense of Bikhchandani et al. (1992); Banerjee (1992). At such points, society stops learning from agents’ private information.

The agent-side welfare term used throughout the paper is therefore the expected loss from choosing the wrong state. It has the following closed form (see Appendix C.1):

$$-C \mathbb{P}(a_i \neq \omega | b_i, q_i) = -C \min(b_i, 1 - b_i, 1 - q_i). \quad (4)$$

3.2 THE PLANNER'S PROBLEM

The planner-side contribution is to study two objectives within the same controlled learning system. The altruistic planner, denoted by subscript A , wants agents to take the correct action ($a_i = \omega$). The biased planner, denoted by subscript B , wants agents to take a specific action, which we set to G , regardless of ω . In both cases the planner controls each private signal's precision, pays a cost governed by a non-negative, increasing, continuous, concave function $\beta(\cdot)$, and observes exactly the same information set $\mathcal{H}_i$ as the agents.

Altruistic Planner For the altruistic planner, the control problem captures costly improvements in informativeness. At time i , the planner chooses q_i and pays $\beta(q_i)$, where $\beta(p) = 0$ for a baseline precision $p \in [0.5, 1]$. Because decreasing precision cannot help an altruistic planner (see Appendix C.9), the notation charges additional cost only for increasing precision above p , with decreasing marginal costs. Agents know both p and $\beta(\cdot)$.

The altruistic objective makes the welfare contribution explicit: maximize agents' expected total utility net of precision costs. Let $r_A(b_i, q_i)$ be the instantaneous reward from public belief b_i after choosing precision q_i for agent i . Starting from b_1 and following policies $\pi = (\pi_i)_{i=1}^\infty$ with $\pi_i(\mathcal{H}_i) = q_i$, the expected discounted utility for discount factor $\delta \in [0, 1]$ is:

$$V_A^\pi(b_1) = \sum_{i=1}^{\infty} \delta^{i-1} r_A(b_i, \pi(b_i)), \quad \text{where} \quad r_A(b_i, q_i) = -\beta(q_i) - C \mathbb{P}(a_i \neq \omega | b_i, q_i). \quad (5)$$

The optimal altruistic value and policy are the supremum and arg supremum of $V_A^\pi(\cdot)$ over policies $\pi \in \Pi$. Setting $\delta = 0$ recovers the myopic benchmark used below, which removes the social-learning externality from the planner's objective.

Biased Planner The biased planner contribution uses the same social-learning machinery but changes the objective and therefore the strategic incentives. As illustrated by the political-campaign example in Appendix A, this planner seeks to induce action G from every agent regardless of ω . If an agent chooses G , the planner incurs cost 0; if the agent chooses B , the planner incurs cost $C > 0$, independent of the state. Unlike the altruistic planner, the biased planner may benefit from either increasing or decreasing precision relative to the baseline p . Any deviation from p requires tailoring information to the agent and costs $\beta(|q_i - p|)$, with $\beta(0) = 0$.

The resulting biased instantaneous reward is:

$$r_B(b_i, q_i) = -\beta(|q_i - p|) - C \mathbb{P}(a_i = B | b_i, q_i) \quad (6)$$

Using Equation (6), the biased value $V_B^\pi(\cdot)$, optimal value $V_B^*(\cdot)$, and optimal policy $\pi_B^*(\cdot)$ are defined analogously to their altruistic counterparts from Equation (5).

This comparison clarifies the alignment stakes of the model. As noted in the first paragraph of Section 3, agents always prefer actions that match ω , so when $\omega = B$, a biased planner who successfully induces G lowers agent utility. The altruistic planner's success always improves agent utility, whereas the biased planner's success is aligned with agents only when $\omega = G$.

For both objectives, the formal payoff is an infinite-horizon discounted stationary Markov Decision Process with state $b_i \in [0, 1]$, control $q_i \in [0.5, 1]$, and transition function Equation (3) (Puterman, 1990). Hence a unique optimal value function exists (Kallenberg, 2011). We focus on deterministic Markov policies $\Pi := \{\pi : [0, 1] \rightarrow [0.5, 1]\}$ with $\pi(b_i) = q_i$, making the policy characterizations in the next sections functions of the public belief alone.

Remark 2. *The framework intentionally imposes three transparency constraints that make the resulting influence claims conservative. (1) The planner has the same history as the agents. This excludes richer planner-side data, but fits settings where all parties rely on the same public signals, such as early findings or survey data. (2) The planner can change signal precision, but signals still*

pass through a binary symmetric channel. This excludes cherry-picking, censorship, and falsification, but fits settings where mediation improves noisy data sources without changing content. (3) The planner's control choices are fully observable. This excludes covert policies and hidden framing, but fits transparent mediation environments.

4 OPTIMAL ALTRUISTIC POLICIES

We begin with the myopic benchmark because it isolates the contribution of social learning in the full dynamic policy. Setting $\delta = 0$ makes the planner ignore all future agents, so $V_A^\pi(b) = r_A(b, \pi(b))$. The myopic altruistic policy $\pi_A^0(\cdot)$ is therefore:

$$\pi_A^0(b) \in \arg \sup_{q \in [0.5, 1]} r_A(b, q) \quad \forall b \in [0, 1] \quad (7)$$

This benchmark also has a decentralized interpretation: each agent chooses and pays for her own signal precision to maximize her expected utility net of cost, with no regard for the information externality imposed on later agents.

Theorem 1. *The optimal myopic altruistic policy π_A^0 is given as follows:*

$$\pi_A^0(b) = \begin{cases} 1 & b \in (t_M, 1 - t_M) \\ p & \text{o.w.} \end{cases} \quad \text{where } t_M = \begin{cases} \frac{\beta(1)}{C} & \beta(1) < C(1 - p) \\ 0.5 & \text{o.w.} \end{cases}$$

Proof in Appendix C.2.

The myopic benchmark gives a scanable threshold contribution. When the public belief is already strong, the myopic planner chooses the zero-cost baseline precision p . When the public belief is weak, the planner provides a perfect signal, $q = 1$, unless that precision is too costly. The threshold depends only on the cost of a perfect signal, $\beta(1)$, and the loss from an incorrect action, C ; if $\beta(1) \geq C$, then a perfect signal is never worth buying, $t_M = 0.5$, and the perfect-signal region is empty.

The main technical contribution for the altruistic planner is the following structural property of the dynamic value function:

Theorem 2. *$V_A^*(\cdot)$ is convex with respect to public belief.*

The proof of Theorem 2 (Appendix C.3) is nonstandard because agents' actions themselves depend on the public belief. If actions were independent of the belief process, as in settings such as Nyarko (1994), expected utility would be linear in the belief state and convexity would follow directly. Here, convexity must be proved while preserving the endogenous social-learning thresholds.

This convexity result is the key step that makes the full optimal altruistic policy characterizable.

Theorem 3. *There exist d_A, t_A such that $0 < d_A \leq t_A \leq t_M \leq 0.5$ and*

$$\pi_A^*(b) = \begin{cases} p & b \in [0, d_A) \cup (1 - d_A, 1] \\ 1 & b \in (t_A, 1 - t_A) \\ \max(b, 1 - b) & \text{o.w.} \end{cases}$$

Furthermore, if $t_M < 0.5$, then $d_A < t_M$. See C.5 for the proof and figure 2a for an example.

The optimal altruistic policy contributes a three-regime view of mediation. First, at extreme public beliefs, the planner stops investing and uses the baseline precision p . This no-investment region is narrower than in the myopic benchmark because the dynamic planner values the effect of the current action on future public beliefs.

Second, when the public belief is close to 0.5 and therefore uninformative, the dynamic optimum selects precision 1 whenever doing so is not cost-prohibitive. By Equation (2), the agent's action then matches the true state with probability 1, causing the public belief to collapse to either 0 or 1.

Third, in the intermediate region, the planner chooses the minimum precision $\max(b, 1 - b)$ that keeps the agent's action responsive to her private signal (see Equation (2)). Any lower precision would make the action uninformative. This regime highlights the social-learning contribution of the policy: the planner pays exactly enough to keep peer observation informative.

5 OPTIMAL BIASED POLICIES

We again begin with the myopic benchmark so that the dynamic contribution of social learning is visible. As in Section 4, the myopic biased policy $\pi_B^0(\cdot)$ maximizes the instantaneous reward $r_B(b, q)$. The biased case introduces a new structural feature: an exact optimum need not exist for every belief (Puterman, 1990, sec. 2.3.1). When this occurs, we state the corresponding ϵ -optimal policies using an ϵ superscript.

Theorem 4. *There exist $t_1, \dots, t_5 \in (0, p)$, $t_1 < 1 - p \leq t_2 \leq t_3 < 0.5 \leq t_4 \leq t_5 < p$ so that:*

- (A) *If $b \in [0, t_1] \cup (1 - p, t_2) \cup [t_3, t_4] \cup (p, 1]$, then $\pi_B^0(b) = p$.*
- (B) *If $b \in (t_1, 1 - p] \cup [t_2, t_3)$, then $\pi_B^0(b) = 1 - b$.*
- (C) *If $b \in (t_4, t_5)$, then $\pi_B^0(b) = 1$.*
- (D) *If $b \in [t_5, p]$, then $\pi_B^0(b)$ does not exist, and $\pi_B^{\epsilon, 0}(b) = b - \epsilon$ for sufficiently small $\epsilon > 0$.*

Proof appears in Appendix C.6. Theorem 4 supplies the myopic structure we use to state the full dynamic characterization for the biased planner.

Theorem 5. *There exist $t_1, t_2 \in [0, p]$ with $t_1 \leq 1 - p \leq 0.5 < t_2 < p$ so that:*

- (A) *If $b \leq t_1$ or $b > p$, then $\pi_B^*(b) = p$.*
- (B) *If $b \in (t_1, 1 - p]$, then $\pi_B^*(b) \geq p$.*
- (C) *If $b \in (1 - p, 0.5)$, then $\pi_B^*(b) \geq 1 - b$.*
- (D) *If $b \in [0.5, t_2)$ and $\pi_B^*(b)$ exists, then $\pi_B^*(b) \geq b$.*
- (E) *If $b \in (t_2, p]$, then $\pi_B^*(b)$ does not exist, and $\pi_B^{\epsilon*}(b) = b - \epsilon$ for sufficiently small $\epsilon > 0$.*

Theorem 5 (proven in Appendix C.8 and depicted in figure 2a) is the main biased-planner contribution: it partitions public beliefs into five strategic phases.

Phase (A) is a cascade-management regime. When the public belief is sufficiently strong, steering may be too costly even if $b < 0.5$ is bad for the planner. The chosen precision p is below $\max(b, 1 - b)$, so Equation (2) implies that the agent follows the public belief regardless of her private signal. The action is uninformative, the public belief stays fixed, and the Markov policy repeats forever: an unfavorable cascade if $b < 0.5$ and a favorable cascade otherwise.

Phase (B) shows how a biased planner uses precision to escape a nearly unfavorable cascade. The planner raises precision enough for the private signal to affect the action even though the signal is expected to be B . This creates a nonzero chance of a favorable action and hence a last-ditch opportunity to move the public belief toward the planner’s preferred outcome.

Phase (C) makes signal obfuscation explicit. For somewhat higher beliefs with $b < 0.5$, more precise signals are more likely to produce unfavorable news for the planner. The planner therefore keeps precision just high enough to affect the agent’s action ($q \geq 1 - b$) while using the least precise signal that can still move the public belief favorably.

Phase (D) is a belief-bolstering regime. Once public belief weakly favors the desired action, $b > 0.5$, higher precision makes the agent more likely to infer $\omega = G$. The planner invests at least at the baseline level to strengthen that favorable belief.

Phase (E) returns to obfuscation, now to protect a favorable public belief. By lowering precision just below $\max(b, 1 - b)$, to $b - \epsilon$, the planner ensures through Equation (2) that agents ignore private signals and choose G . The risk that a private signal would overturn the favorable belief outweighs both the cost of reducing precision and the possible benefit of further increasing the public belief.

6 EVALUATION VIA LLM-BASED SIMULATION

The empirical contribution of this section is to test whether the analytical policy structure remains informative when the planner and agents are instantiated by LLMs. The evaluation is organized

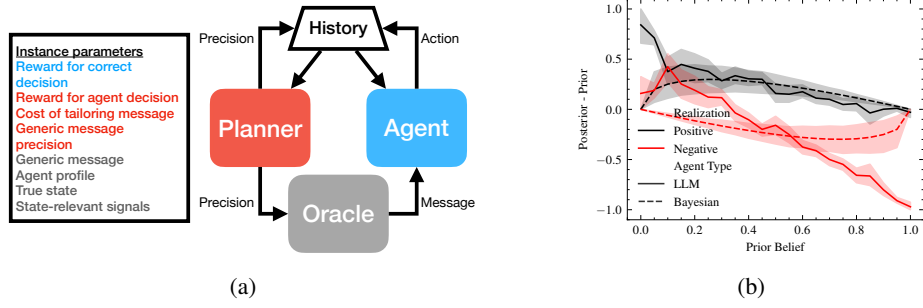


Figure 1: Experimental setup and agent-belief validation. (a) The LLM simulation instantiates the theoretical framework with distinct agent, planner, and oracle roles; color coding indicates which instance parameters each role observes. (b) LLM belief updates (solid) deviate from Bayesian updates (dashed) after positive (black) and negative (red) signals at a fixed prior.

around three questions that mirror the paper’s theory: how LLM agents depart from Bayesian updating, whether an LLM planner recovers the predicted belief-dependent policy regimes, and how those policies affect social welfare.

We operationalize controlled social learning through a car-purchase scenario in which agents decide whether to buy a new car. As shown in figure 1a, LLMs play three roles, with prompts provided in Appendix E.2.2: **Agent**: role-playing an assigned profile, the agent observes previous actions and a private message about the car with known precision, reports a belief about the car’s quality, and chooses whether to buy.

Planner: after observing the action history, the planner chooses the precision q_i of agent i ’s private signal according to its objective.

Oracle: given the planner’s chosen precision, an agent profile, and a car fact sheet (Appendix E.2.1), the oracle generates a tailored private signal with the requested precision. We compare these LLM simulations with numerical evaluations of the planner’s MDP. In both experimental modes, we use a linear cost function, $\beta(q) = k|q - p|$, vary k , baseline precision p , and discount factor δ , and fix the cost of an incorrect action at $C = 1$. Appendix E gives the full setup and prompts, and Appendix E.3 validates both the LLM belief reports and the oracle’s ability to produce messages with the requested precision.

The experiments contribute three empirical findings:

1. **Social-learning control materially changes outcomes.** Analytical and LLM planners improve their own utility by accounting for how current actions affect future public beliefs; depending on objective alignment, the same mechanism can improve or reduce social welfare.
2. **The analytical policy structure is predictive beyond Bayesian agents.** Even when facing non-Bayesian LLM agents, LLM planners choose policies that closely resemble the non-obvious optimal regimes characterized for altruistic planners in Section 4 and for biased planners above.
3. **LLM planners exhibit emergent strategic behavior.** The LLM planner not only approximates the analytical policy, but also deviates in ways that appear adapted to the non-Bayesian updating patterns of LLM agents documented in Section 6.1.

6.1 LLMs AS NON-BAYESIAN AGENTS

The first empirical step is to identify which agent behavior the planner must adapt to. Although the analytical model assumes Bayesian agents, humans and LLMs both exhibit systematic cognitive biases (Rupprecht et al., 2025). We therefore isolate LLM belief updating by giving an LLM a prior and a new signal, then comparing its reported posterior to the Bayesian update. The resulting deviations, shown in figure 1b and further illustrated in Appendix E.4, are summarized by three patterns:

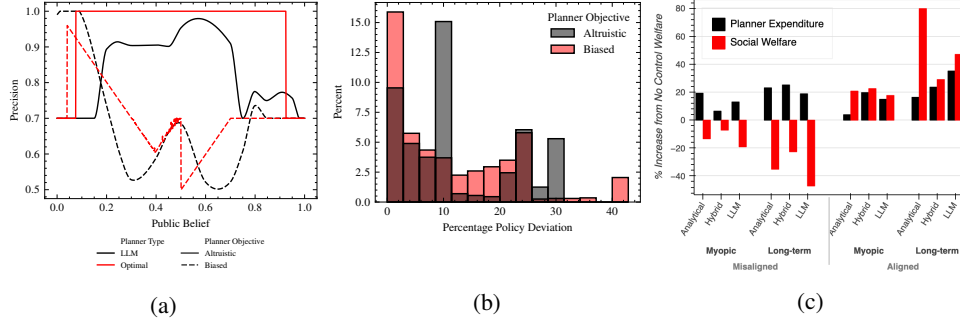


Figure 2: Main empirical evidence for the paper’s theoretical contributions. (a) LLM planner policies (black) and analytically optimal policies (red) show similar belief-dependent regimes for altruistic (solid) and biased (dashed) objectives. (b) The percentage deviation between LLM and optimal policies is concentrated near small values. (c) Planner expenditure and social-welfare change, reported as a percent of the no-control baseline welfare, show the welfare consequences of biased and altruistic control when the true state is fixed to B.

(NB1) LLM agents underreact to private signals which align with their prior beliefs.

(NB2) LLM agents overreact to private signals which run counter to their prior beliefs.

(NB3) As a result, LLM agents require a stronger public belief to enter an information cascade.

These patterns matter for the paper’s contribution because they test whether the theoretical policy regimes survive behavioral misspecification. The same biases have been observed empirically in human studies (e.g., Ba et al. (2022); Chan et al. (2025)), and the next sections show that the LLM planner’s policy remains structurally close to the analytical characterization while adapting to these non-Bayesian agents.

6.2 VALIDATION OF PLANNER POLICY STRUCTURE AND THE ROLE OF SOCIAL LEARNING

We next evaluate the central policy-structure contribution by comparing LLM planner policies with the optimal policies from Sections 4 and 5. As shown in figure 2a, the emergent LLM strategies are structurally similar to the theoretical optima despite non-Bayesian agent behavior. This is especially salient in the biased case, where the optimal policy is nontrivial: the LLM planner appears to exploit the same social-learning logic of escaping unfavorable beliefs, protecting favorable beliefs, and modulating signal precision to control whether actions reveal private information via Equation (2).

The qualitative regimes align across planner types. In the altruistic case, both planners stop investing when public belief is strong and invest heavily when public belief is uncertain. In the biased case, both planners display high investment to escape an unfavorable belief, reduced precision near the midpoint, and no investment once a favorable belief is established. Figure 2b shows that the deviation is less than 10% for the majority of belief states for both altruistic and biased planners, supporting the claim that the analytical characterization captures the main structure of the emergent LLM strategy.

The differences are also informative because they reveal how LLM planners adapt the theoretical regimes to LLM agents. (1) The LLM planner avoids extreme precisions (0.5 or 1.0), consistent with central tendency bias (Rupprecht et al., 2025). (2) The policy tapers investment more gradually than the analytical cutoff, reflecting the agents’ resistance to cascades (NB3); the planner appears to treat strong public beliefs as less final because countervailing private signals can still move agents. (3) The biased LLM planner continues investing even at very low beliefs, consistent with exploiting possible overreaction (NB2) to a surprisingly positive signal. These deviations preserve the contribution’s main message: the LLM planner uses social-learning structure, but its exact policy reflects the behavior of the agents it faces.

6.3 WELFARE IMPLICATIONS OF ALTRUISM, BIAS, AND ALIGNMENT

The final empirical contribution is to quantify the welfare stakes of controlled social learning. We define social welfare $W^\pi(b)$ as the total expected discounted utility of all agents. Because better information never lowers an agent’s expected utility, social welfare is monotonic in signal precision (see Appendix C.9), so the altruistic planner increases welfare relative to the baseline. We compare welfare and planner expenditure in three settings: (1) the *analytical* setting with the optimal policy and Bayesian agents, (2) the *LLM* setting with an LLM planner and LLM agents, and (3) a *hybrid* setting with the optimal policy applied to LLM agents.

The results in figure 2c show that controlled information mediation can substantially alter social welfare. They also show why accounting for social learning is essential: myopic policies, which ignore future belief effects, substantially worsen planner outcomes. When misaligned, the biased analytical and LLM planners reduce social welfare by 40 to 50%. This reduction is achieved through policies that intentionally obscure information about the true state by decreasing signal precision. The magnitude is notable given the transparency constraints in Remark 2, and it substantiates the risk posed by potentially misaligned LLM information mediators.

The welfare comparison also surfaces a behavioral-modeling lesson. In the hybrid setting, the analytically optimal policy is designed for Bayes-rational agents and is therefore brittle when applied to non-Bayesian LLM agents. The LLM policy, by contrast, remains close to the analytical policy structure while better matching the human-like biases of the agents it controls.

7 CONCLUSION

This paper contributes a framework, theory, and LLM-based evaluation for strategic control of social learning by algorithmic information mediators. The framework shows how a planner can steer a public belief by choosing the precision of private signals while agents retain decentralized decision authority. The theory characterizes optimal altruistic and biased policies, including regimes of investment, non-investment, and intentional precision reduction. The simulations show that these mechanisms can substantially change social welfare and that LLM planners can recover much of the analytical policy structure while adapting to non-Bayesian LLM agents.

The main takeaway is that LLM-mediated influence should be evaluated as a social process, not only as a collection of individual interactions. First, human-AI or human-algorithm interaction can be mischaracterized if the analysis ignores social learning and the resulting information externalities. Second, even constrained information mediators that do not lie, censor, or hide their controls can guide or derail public beliefs. Third, modern LLMs can exhibit strategic behavior that accounts for social learning and for non-Bayesian cognitive patterns. A limitation of our study is the absence of human-subject data. Because the fidelity of LLM-human simulation remains contested (see, e.g., Horton (2023); Gao et al. (2025)), human data would be valuable for validating or fine-tuning the LLM simulators.

Two future directions follow directly from the contribution structure. The first is to generalize the model and results. Larger state spaces, richer signal structures, more general loss functions, and heterogeneous agents would bring the framework closer to real information ecosystems. Extending beyond binary states and 0-1 losses appears primarily algebraic, and we conjecture that the qualitative policy regimes will persist. More general signal structures pose a sharper theoretical challenge because the convexity result of Theorem 2 may fail; identifying the broadest class of signal structures that preserves it is an important open problem. As a first heterogeneous-agent step, Appendix D generalizes the altruistic planner results to agents with opposing preferences.

The second direction is mitigation. Because biased planners in our model can reduce welfare even under strong transparency constraints, future work should study regulations or mechanisms that better align planner and agent incentives and limit harmful control of social learning.

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A APPLICATIONS ELABORATED

This appendix makes the two planner objectives concrete. For an altruistic-planner application, consider agents deciding whether to patronize a business or service provider, such as a restaurant, contractor, or realtor, while a recommendation system supplies information about that business. The binary state is whether the business is good, and it is unknown to both agents and the system. Agents want to patronize the business if and only if it is good, and the recommendation system wants to induce exactly those correct decisions. In this sense, the recommendation system is an *altruistic planner*.

The recommendation system controls precision by choosing between highly targeted ads and generic or confusing ones. A targeted, *precise* ad presents strengths and weaknesses in a context that matches the agent’s background, making it more informative, comprehensible, and relatable. Such a signal is more likely to lead the agent to the correct action, so social welfare rises as more agents make state-matching decisions. Tailoring precision is costly, however, because it requires adapting content to the agent’s background. The altruistic planner’s application-level problem is therefore to choose precisions that maximize social welfare net of tailoring costs.

For a biased-planner application, consider voters who are motivated to support the candidate most likely to win in their community. This motive to “back a winner” can affect electoral outcomes; for example, it helps explain why US states with early primary elections have outsized influence (see Bartels (1988)). The planner is one candidate’s campaign, so it wants agents to back that candidate. The binary state is whether the candidate is winning. If an agent knew the state, she would back the candidate if and only if he were winning.

The campaign controls precision through the format and clarity of political information. An agent who understands her community can infer whether the candidate is winning if she accurately understands his stances and policies. That understanding depends on presentation: some agents process audio-visuals best, others prefer long-form articles with facts and citations, others need brief focused content, and others require their native language.

In this application, the campaign sends digital ads, modeled as private signals, with varying precision. Precision captures how clearly and effectively the ad conveys information tailored to the agent’s taste. A high-precision signal helps the agent understand the candidate’s policies, track record, and character, enabling her to infer whether the candidate will win. This does not necessarily help the planner: if the candidate is losing in the agent’s community, a precise signal would push the agent against the planner’s candidate. A low-precision signal can instead obfuscate the state, for example by being vague or confusing, and thereby raise the chance that the agent believes the candidate is winning. This example makes the biased-planner contribution concrete: when the planner believes its candidate is losing, it may strategically decrease precision. Tailoring either precise or imprecise ads is costly because the planner must know how the agent processes content; even confusing an agent effectively may require knowing, for example, that she prefers brief messages and can be confused by verbose articles. The planner therefore chooses precisions to maximize expected backers net of tailoring costs.

B SOCIAL LEARNING DYNAMICS

For completeness, we restate the controlled social-learning timeline used in the main model. At step i :

1. The planner determines q_i based on $\mathcal{H}_i$ and, q_i is observed by all agents.
2. Agent i receives private signal s_i such that $\mathbb{P}\{s_i = \omega\} = q_i$.
3. Agent i takes action a_i based on the observable history and their private signal with the aim of maximizing their own utility.
4. All other agents (and the planner) observe a_i and the public belief at the end of step i , b_{i+1} is updated accordingly.

Because step (1) makes q_i a function of $\mathcal{H}_i$, conditioning on both q_i and $\mathcal{H}_i$ makes the conditioning on q_i redundant. We use this simplification throughout the appendix derivations.

B.1 PRIVATE BELIEF UPDATE

This subsection derives the private belief formula used to obtain the main text's action thresholds. Let $\alpha_i := \frac{b_i}{1-b_i}$ be the public likelihood ratio implied by observed actions. Agent i combines this public belief with private signal s_i to form $\tilde{b}_i = \mathbb{P}\{\omega = G | \mathcal{H}_i, q_i, s_i\}$ and corresponding private likelihood ratio $\tilde{\alpha}_i := \frac{\tilde{b}_i}{1-\tilde{b}_i}$. The following cases compute $\tilde{\alpha}_i$ and $\tilde{b}_i$.

1. If private signal is $s_i = G$, then

$$\begin{aligned} \tilde{\alpha}_i &= \frac{\tilde{b}_i}{1-\tilde{b}_i} = \frac{\mathbb{P}\{\omega = G | \mathcal{H}_i, q_i, s_i = G\}}{\mathbb{P}\{\omega = B | \mathcal{H}_i, q_i, s_i = G\}} \\ &= \frac{\mathbb{P}\{s_i = G | \omega = G, \mathcal{H}_i, q_i\} \mathbb{P}\{\omega = G | \mathcal{H}_i, q_i\}}{\mathbb{P}\{s_i = G | \omega = B, \mathcal{H}_i, q_i\} \mathbb{P}\{\omega = B | \mathcal{H}_i, q_i\}} \\ &= \frac{q_i}{1-q_i} \frac{b_i}{1-b_i} \end{aligned} \quad (8)$$

$$= \frac{q_i}{1-q_i} \alpha_i \quad (9)$$

Equation (8) uses the conditional-independence assumption: given ω , agents' signals are independent of one another and of the history, so $\mathbb{P}\{s_i = G | \omega = G, \mathcal{H}_i\} = \mathbb{P}\{s_i = G | \omega = G\} = q_i$. The same argument applies under $\omega = B$, and we also use that q_i is determined by $\mathcal{H}_i$.

2. If private signal is $s_i = B$, following the logic above,

$$\tilde{\alpha}_i = \frac{1-q_i}{q_i} \alpha_i \quad (10)$$

To express the posterior in belief-state notation, let $y(b, q)$ and $z(b, q)$ denote the probabilities of signals G and B under public belief b and precision q . Conditioning on ω and applying the law of total probability gives:

$$\begin{aligned} y(b_i, q_i) &= \mathbb{P}\{s_i = G | \mathcal{H}_i\} \\ &= \sum_{k \in \{G, B\}} \mathbb{P}\{s_i = G | \mathcal{H}_i, \omega = k\} \mathbb{P}\{\omega = k | \mathcal{H}_i\} \\ &= q_i b_i + (1-q_i)(1-b_i) = 1 + 2q_i b_i - q_i - b_i \end{aligned} \quad (11)$$

Similarly,

$$z(b_i, q_i) = q_i + b_i - 2q_i b_i \quad (12)$$

Combining Equation (9), Equation (10), Equation (11), and Equation (12) yields the following:

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = f(b_i, q_i, s_i) = \begin{cases} \frac{q_i}{y(b_i, q_i)} b_i & s_i = G \\ \frac{1-q_i}{z(b_i, q_i)} b_i & s_i = B \end{cases} \quad (13)$$

The expression is well-defined: if $s_i = G$, then $y(b_i, q_i) = \mathbb{P}(s_i = G | \mathcal{H}_i)$ cannot be 0, and if $s_i = B$, then $z(b_i, q_i)$ cannot be 0.

B.2 AGENT ACTIONS AND PUBLIC BELIEF

We next derive the action rule that determines when an observed action is informative. Agent i chooses G if $\tilde{b}_i > 0.5$ (equivalently, $\tilde{\alpha}_i > 1$), chooses B if $\tilde{b}_i < 0.5$ (equivalently, $\tilde{\alpha}_i < 1$), and follows s_i under the tie-breaking rule. The action and precision q_i are observed by all peers and determine the next public belief.

When $b_i < 1 - q_i$, we have $\alpha_i < \frac{1-q_i}{q_i}$. If $s_i = G$, then $\tilde{\alpha}_i < 1$ by Equation (9); if $s_i = B$, then $\tilde{\alpha}_i < \frac{(1-q_i)^2}{q_i^2} \leq 1$ because $q_i \geq 0.5$, by Equation (10). Thus $a_i = B$ for every signal realization when $b_i < 1 - q_i$. A symmetric argument gives $a_i = G$ when $b_i > q_i$. These are the cascade regions in which the public belief overpowers the private signal.

In the remaining interval, $1 - q_i \leq b_i \leq q_i$, the private signal is decisive. If $s_i = G$, then $\tilde{\alpha}_i \geq 1$ by Equation (8) and $b_i \geq 1 - q_i$; if $s_i = B$, then $\tilde{\alpha}_i \leq 1$ by Equation (10) and $b_i \leq q_i$. With the tie-breaking rule, $a_i = s_i$, yielding Equation (2), reproduced below.

$$a_i = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & q_i < 1 - b_i \end{cases}$$

Therefore, the action is informative only in the first case of Equation (2), where future agents can perfectly infer agent i 's private signal. In that case, the public belief updates exactly as the private belief in Equation (13). In the other two cases, the action adds no information about the state and the public belief remains unchanged. This gives Equation (3), reproduced below:

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (14)$$

B.3 MARKOVIANITY AND SUFFICIENCY OF PUBLIC BELIEF

This subsection proves the sufficiency claim used to formulate the planner's MDP: $\mathbb{P}\{a_i = a | \mathcal{H}_i, q_i\}$ depends on the history only through b_i and q_i for $a \in \{G, B\}$. Hence,

$$\mathbb{P}(a_i = a | \mathcal{H}_i, q_i) = \mathbb{P}(a_i = a | b_i, q_i) \quad (15)$$

From Equation (2), for the second and third cases, it is clear that $\mathbb{P}\{a_i = a | \mathcal{H}_i, q_i\}$ is a function of b_i, q_i for $a \in \{G, B\}$, regardless of the rest of $\mathcal{H}_i$. We consider the first case next.

$$\begin{aligned} & \mathbb{P}(a_i = G | \mathcal{H}_i, q_i) \\ &= \sum_{w \in \{G, B\}} \mathbb{P}(\omega = w | \mathcal{H}_i, q_i) \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = w) \\ &= b_i \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = G) \\ &\quad + (1 - b_i) \mathbb{P}(a_i = G | \mathcal{H}_i, q_i, \omega = B) \\ &= b_i \mathbb{P}(s_i = G | b_i, q_i, \omega = G) \\ &\quad + (1 - b_i) \mathbb{P}(s_i = G | b_i, q_i, \omega = B) \\ &= b_i q_i + (1 - b_i)(1 - q_i) \end{aligned} \quad (16)$$

The first step applies the law of total probability. The second uses that q_i is a function of $\mathcal{H}_i$, allowing substitution of b_i for $\mathbb{P}(\omega = w | \mathcal{H}_i, q_i)$. The third uses $a_i = s_i$ when $1 - q_i \leq b_i \leq q_i$ and the conditional independence of s_i from the history given ω .

This establishes the Markovianity statement:

Theorem 6. *Markovianity of the Public Belief Process*

$$b_{i+1} = \mathbb{P}\{\omega = G | b_i, q_i, a_i\} \quad (17)$$

Thus b_i captures all information in $\mathcal{H}_i$ that is relevant for updating b_{i+1} .

Proof. Note that, from step (1) in Appendix B, q_i is a function of $\mathcal{H}_i$. It suffices to show

$$b_{i+1} = \frac{\mathbb{P}\{a_i \cap \omega = G | b_i, q_i\}}{\mathbb{P}\{a_i | b_i, q_i\}}$$

We will prove the claim via induction on i .

For our base case, consider $i = 1$ so that $\mathcal{H}_2 = \{b_1, (q_1, a_1)\}$ and $b_i = b_1$. An application of Bayes' Theorem provides the following:

$$\begin{aligned} b_2 &= \frac{\mathbb{P}\{\omega = G|b_1, q_1\} \mathbb{P}\{a_1|\omega = G, b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\}} = \frac{b_1 \mathbb{P}\{a_1|\omega = G, b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\}} \\ &= \frac{b_1 \mathbb{P}\{a_1 \cap \omega = G|b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\} \mathbb{P}\{\omega = G|b_1, q_1\}} \\ &= \frac{\mathbb{P}\{a_1 \cap \omega = G|b_1, q_1\}}{\mathbb{P}\{a_1|b_1, q_1\}} \end{aligned}$$

where the final step follows as the unconditioned belief of ω is equal to the prior belief b_1 .

Now, assume the claim holds for all $i \leq n$.

$$\begin{aligned} b_{n+1} &= \mathbb{P}\{\omega = G|\mathcal{H}_n, a_n, q_n\} = \frac{\mathbb{P}\{\omega = G|\mathcal{H}_n, q_n\} \mathbb{P}\{a_n|\omega = G, \mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n|\mathcal{H}_n, q_n\}} \\ &= \frac{b_n \mathbb{P}\{a_n|\omega = G, \mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n|\mathcal{H}_n, q_n\}} \\ &= \frac{b_n \mathbb{P}\{a_n \cap \omega = G|\mathcal{H}_n, q_n\}}{\mathbb{P}\{a_n|b_n, q_n\} \mathbb{P}\{\omega = G|\mathcal{H}_n, q_n\}} \\ &= \frac{\mathbb{P}\{a_n \cap \omega = G|b_n, q_n\}}{\mathbb{P}\{a_n|b_n, q_n\}} \end{aligned}$$

where the final two steps follow from Equation (15) and the definition of b_n .

This completes the proof. $\square$

This proves the Markovian public-belief representation by adapting the arguments of Banerjee (1992) and Bikhchandani et al. (1992) to the controlled setting, where a central planner selects agents' signal precisions.

C PROOFS

The proof appendix supports the main contribution structure by making explicit which technical facts deliver each policy characterization. We first derive the closed-form mistake probability used in both planner objectives, then prove the myopic benchmarks, the convexity result for the altruistic value function, and the optimal-policy characterizations for altruistic and biased planners.

C.1 DERIVATION OF EQUATION (4)

Lemma 7.

$$\mathbb{P}(a_i \neq \omega|b_i, q_i) = \min(b_i, 1 - b_i, 1 - q_i)$$

Proof. We prove this for each case of a_i delineated in Equation (2). In the second case, we can reason as follows:

$$\mathbb{P}(a_i \neq \omega|b_i, q_i) = \mathbb{P}(\omega \neq G|b_i, q_i) = 1 - b_i$$

Similarly, in the third case of Equation (2), we obtain:

$$\mathbb{P}(a_i \neq \omega|b_i, q_i) = \mathbb{P}(\omega \neq B|b_i, q_i) = b_i$$

For the case when $1 - q_i \leq b_i \leq q_i$ and $a_i = s_i$ from Equation (2), we can reason as follows:

$$\begin{aligned} \mathbb{P}(a_i \neq \omega | b_i, q_i) &= \sum_{a \in \{G, B\}} \mathbb{P}(a_i = a | b_i, q_i) \mathbb{P}(\omega \neq a | b_i, q_i, a_i = a) \\ &= \mathbb{P}(s_i = G | b_i, q_i) \mathbb{P}(\omega \neq G | b_i, q_i, s_i = G) \\ &\quad + \mathbb{P}(s_i = B | b_i, q_i) \mathbb{P}(\omega \neq B | b_i, q_i, s_i = B) \end{aligned} \quad \text{Eqn 2} \quad (18)$$

$$\begin{aligned} &= y(b_i, q_i) \mathbb{P}(\omega = B | b_i, q_i, s_i = G) \\ &\quad + z(b_i, q_i) \mathbb{P}(\omega = G | b_i, q_i, s_i = B) \end{aligned} \quad \text{Eqns 11, 12, Thm 6} \quad (19)$$

$$= \begin{cases} y(0, q_i) & b_i = 0 \\ y(b_i, q_i)(1 - \frac{b_i q_i}{y(b_i, q_i)}) + z(b_i, q_i) \frac{b_i(1-q_i)}{z(b_i, q_i)} & b_i \neq 0 \end{cases} \quad \text{Eqn 13} \quad (20)$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ y(b_i, q_i) - b_i q_i + b_i(1 - q_i) & b_i \neq 0 \end{cases}$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ y(b_i, q_i) - b_i q_i + b_i - b_i q_i & b_i \neq 0 \end{cases}$$

$$= \begin{cases} 1 - q_i & b_i = 0 \\ 1 - q_i & b_i \neq 0 \end{cases}$$

Combining the three cases yields the claim. $\square$

C.2 PROOF OF THEOREM 1: MYOPIC ALTRUISTIC POLICY

Applying Equation (4) to $r_A(b, q)$ in Equation (5) allows us to reason as follows

$$\frac{\partial r_A(b, q)}{\partial q} = \begin{cases} -\dot{\beta}(q) & q < \max(b, 1 - b) \\ -\dot{\beta}(q) + C & q \geq \max(b, 1 - b) \end{cases} \quad (21)$$

As stated in Section 3.2, we can restrict the altruistic planner to precision $q \in [p, 1]$ because social welfare is increasing in precision (see Appendix C.9). Accordingly, $\beta(p) = 0$.

Because $\beta(\cdot)$ is increasing, the first case of Equation (21) leads to optimal precision p if $p < \max(b, 1 - b)$. That is, if the signal is not precise enough to influence the agent's action, then the planner will not expend resources for it. In the latter case, because $\beta(\cdot)$ is increasing and concave, $\dot{\beta}(\cdot)$ is positive and decreasing. Thus, $r_A(b, q)$, if the optimal lies in this range it must be attained at $q = 1$.

We now identify the regime in which the latter case is optimal.

$$\begin{aligned} r_A(b, p) &< r_A(b, 1) \\ -C \min(b, 1 - b, 1 - p) &< -\beta(1) \\ \min(b, 1 - b, 1 - p) &> \frac{\beta(1)}{C} \end{aligned}$$

If $1 - p \leq \frac{\beta(1)}{C}$, the above inequality is never satisfied, and it is never optimal to select precision 1. Otherwise, the inequality above yields $b \in \left[\frac{\beta(1)}{C}, 1 - \frac{\beta(1)}{C} \right]$. This yields Theorem 1. $\blacksquare$

C.3 PROOF OF THEOREM 2: ALTRUISTIC VALUE FUNCTION CONVEXITY

Proof Sketch We inductively prove that the expected k -th stage reward, i.e., the expected utility of the planner from the control and action of the k -th agent, is convex. The instantaneous reward (Equation (5)), which is convex with respect to public belief, provides our base case.

The first challenge encountered is the unusual nature of the public belief update. Although the state space is uncountably infinite, the belief update only takes support on a maximum of 2 values. To manage this, we define a decision tree, i.e., the complete binary tree of all possible trajectories once an initial belief and policy are fixed. Each node has two children corresponding to each possible signal realization the next agent might receive. The root is the expected instantaneous reward at time 1, i.e., from the first agent's action. The induction moves down the levels of this tree with the k -th level containing 2^{k-1} nodes, each associated with a sequence of realizations of $k - 1$ signals.

We then show that for a node in the $(k - 1)$ -th level that has convex expected reward, its two children in the k -th level satisfy the same property. This is where we must deal with the dependence of agents' actions on public belief. Note that even when applying the same precision and receiving the same signal realization, two agents beginning at different public beliefs may take opposing actions (see Equation (2)). Thus, standard results that provide easy ways of bounding the future terms of the Markov process do not apply. Here, our specific belief update is actually helpful. We can leverage the fact that Bayesian updates are martingales (i.e., $\mathbb{E}[b_{i+1}] = b_i$). Along with the convexity of instantaneous rewards, this allows us to complete the inductive step and, subsequently, the proof.

Preliminaries We first prove the following lemma containing a few useful properties of the optimal value function $V_A^*(\cdot)$.

Lemma 8.

$V_A^*(\cdot)$ is non-positive, $V_A^*(b) = V_A^*(1 - b) \forall b \in [0, 1]$, and $V_A^*(0) = V_A^*(1) = 0$ with $\pi_A^*(0) = \pi_A^*(1) = p$.

Proof.

Non-positivity

$$r(b_i, q_i) \leq 0 \quad \forall b_i, q_i \rightarrow V_A^* \leq 0$$

Symmetry

Let $n_i = 1 - b_i$ and let Λ_n and Λ_b be the decision trees rooted at states n_0 and b_0 , respectively. Showing that Λ_n and Λ_b are symmetric suffices to show the symmetry of $V^*(\cdot)$. This will be accomplished by showing that the expected rewards, belief state values, and branching probabilities of the trees are reflections of one another.

First consider the instantaneous reward function r_A . Note that the decision function of the agents when choosing their actions is symmetric from Equation (2), thus, $\mathbb{P}\{a_i = G|b_i\} = \mathbb{P}\{a_i = B|n_i\}$. Applying this, we obtain

$$\begin{aligned} r(n_i, q_i) &= -\beta(q_i) - Cn_i \mathbb{P}\{a_i = B|\omega = G, n_i\} - C(1 - n_i) \mathbb{P}\{a_i = G|\omega = B, n_i\} \\ &= -\beta(q_i) - C(1 - b_i) \mathbb{P}\{a_i = B|\omega = G, n_i\} - Cb_i \mathbb{P}\{a_i = G|\omega = B, n_i\} \\ &= -\beta(q_i) - C(1 - b_i) \mathbb{P}\{a_i = G|\omega = B, b_i\} - Cb_i \mathbb{P}\{a_i = B|\omega = G, b_i\} \\ &= r(b_i, q_i) \end{aligned}$$

Thus, the instantaneous reward function is symmetric about 0.5.

Now consider the belief state update when beginning with belief n_i assuming signal precision q_i . we will use $\bar{x}$ and $\underline{x}$ to represent updated beliefs from x after observing action G or B , respectively. From Equation (13):

$$\begin{aligned} \bar{n}_i &= \frac{q_i}{y(n_i, q_i)} n_i = \frac{q_i}{z(b_i, q_i)} (1 - b_i) = 1 - \frac{1 - q_i}{z(b_i, q_i)} b_i = 1 - \underline{b}_i \\ \underline{n}_i &= \frac{1 - q_i}{z(n_i, q_i)} n_i = \frac{q_i}{y(b_i, q_i)} (1 - b_i) = 1 - \frac{q_i}{y(b_i, q_i)} b_i = 1 - \bar{b}_i \end{aligned}$$

Thus, the belief updates are also symmetric. Combining this with the symmetry of the reward we obtain:

$$\begin{aligned} r(\bar{n}_i) &= r(1 - \underline{b}_i) = r(\underline{b}_i) \\ r(\underline{n}_i) &= r(1 - \bar{b}_i) = r(\bar{b}_i) \end{aligned}$$

Thus, the expected rewards and belief state values of Λ_n and Λ_b are mirror images of one another.

Finally, consider the branching probabilities which we will denote as $y(b, q) = 1 + 2bq - b - q$ for observing action G and $z(b, q) = 1 - y(b, q)$ for observing action B .

$$\begin{aligned} y(n, q) &= 1 + 2nq - n - q = 1 + 2(1 - b)q - (1 - b) - q = b + q - 2bq = z(b, q) \\ z(n, q) &= 1 - y(n, q) = 1 - z(b, q) = y(b, q) \end{aligned}$$

Thus, the branching probabilities of Λ_n and Λ_b are also symmetric, completing the proof.

Extremal values

We will show that $V_A^*(0) = 0$ and rely on the symmetry about 0.5 proven above to show that $V_A^*(1) = 0$.

From Equation (3), when $b_0 = 0$, $b_0^+ = b_0^- = 0$ for any value of precision. Thus, regardless of signal realizations or control actions, the agents' actions and, subsequently, the public belief, will remain unchanged. That is, $b_i = 0 \forall i$ and, from Equation (2), $a_i = B \forall i$.

From Equation (5), we can then write:

$$\begin{aligned}
V_A^*(0) &= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r(b_i, \pi(b_i)) \right] \\
&= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r(0, \pi(0)) \right] \\
&= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i [-\beta(\pi(0)) - C b_i \mathbb{P}\{a_i = B | \omega = G, b_i = 0\} - C(1 - b_i) \mathbb{P}\{a_i = G | \omega = B, b_i = 0\}] \right] \\
&= \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i [-\beta(q_i)] \right]
\end{aligned} \tag{22}$$

Thus, to attain $V_A^*(0)$, it suffices to choose $\pi(0)$ such that $\beta(\pi(0))$ is minimized for each i . Recall that $\beta(\cdot)$ is increasing on $[0.5, 1]$ and takes its minimum at $\beta(p) = 0$. Thus, by choosing $q_i = p \forall i$, Equation 22 yields $V_A^*(0) = 0$ with corresponding optimal policy $\pi_A^*(0) = p$.

The aforementioned symmetry about 0.5, shows that $V_A^*(1) = 0$, completing the proof. $\square$

We now introduce the concept of *decision trees* for Markovian and more general policies. A policy can be represented by a decision tree, which is a complete binary tree as in figure 3. Every node of the decision tree represents the state of the system, i.e., the corresponding belief value when the controller sends its private signal to the agent corresponding to the level of the node. The branches represent the actions taken by the agent corresponding to the level once it receives its private signal. The controller's policies (i.e. choice of precision) determine the probabilities of the actions from each node and the belief resulting from the action. The probabilities of an action at a certain node can be considered the weight of the branch. A decision tree is uniquely identified by the belief values associated with nodes and the weights of the branches. For a deterministic stationary Markovian policy, the controller's choice of precision, and therefore the probability associated with an action, is a deterministic function of the belief value of the node, regardless of the level of the node in the tree and the path to the node from the root. Thus, the decision tree of any such policy, i.e. the belief values at the nodes and the weights of the branches, is determined entirely by the value of the initial state. For an arbitrary policy of the controller, the controller's decision at a given node can depend on the entire path leading to the node.

Proof of Theorem 2 We now prove the convexity of $V_A^*(\cdot)$ with respect to public belief.

Proof.

We will first show that, for fixed precision, the instantaneous reward r_A is convex in public belief. We can rewrite r_A as follows using Lemma 7:

$$r_A(x, q) = -\beta(q) - C \min(x, 1 - x, 1 - q) \tag{23}$$

The pointwise minimum of linear functions is concave. As we are subtracting the minimum, the overall function is convex with respect to x .

We consider arbitrary x_0, m_0, n_0 such that $tm_0 + (1 - t)n_0 = x_0$ for some $t \in [0, 1]$. To show the convexity of $V_A^*(\cdot)$ we must show that $V_A^*(x_0) \leq tV_A^*(m_0) + (1 - t)V_A^*(n_0)$. By definition of the optimal value function, $V_A^*(x) \geq V_A^\pi(x)$ for any policy π . Thus, it will suffice to show that there exist $\tilde{\pi}$ such that

$$V_A^*(x_0) \leq tV_A^{\tilde{\pi}}(m_0) + (1 - t)V_A^{\tilde{\pi}}(n_0) \tag{24}$$

We refer to the decision tree of π^* when the initial state is x_0 as Λ .

We consider $\tilde{\pi}$ to be a policy of the controller that chooses the precision at each node to be the same as what Λ does at the counterpart node. The nodes of decision trees are considered to be counterparts if they can be reached via the same sequence of actions from the root (see figure 3 for illustration). Thus, the precision chosen by this policy is the same as what Λ does after the same set of actions of the agents. We refer to the decision trees for $\tilde{\pi}$ with initial state m_0 (respectively, n_0) as Λ_m (Λ_n).

We now illustrate the policy $\tilde{\pi}$ for initial values m_0 and n_0 considering the decision trees Λ_m and Λ_n . While doing so, we illustrate counterpart nodes and the weights of the branches. We will select one path of actions λ out of these decision trees by specifying set $\mathcal{G}$ as the epochs where action G is taken and $\mathcal{B}$ as its complement. Refer to figure 3 for illustration. Considering the decision tree Λ , we will refer to x_i as the state at epoch i when beginning with belief x_0 and following π^* along this path of actions. That is, x_i assumes one of 2^i possible belief values at epoch i starting from x_0 . In decision tree Λ_m (respectively, Λ_n), we use m_i (n_i) to denote the

public belief after beginning with m_0 (n_0), following the path λ . Node m_i (respectively n_i) in decision tree Λ_m (Λ_n) is the counterpart of node x_i from decision tree Λ . Note that, $\tilde{\pi}(m_i)$ can now be illustrated notationally. Specifically, $\tilde{\pi}(m_i) = \tilde{\pi}(n_i) = \pi_A^*(x_i)$. Similarly, $\tilde{\pi}$ can be defined at each node of Λ_m , considering the action taken by π^* at each counterpart node in Λ . Note that $y(x_i, \pi_A^*(x_i))$ (respectively, $z(x_i, \pi_A^*(x_i))$) is the weight of the branch corresponding to G (B) emanating from x_i in Λ . Because $\tilde{\pi}(m_i) = \tilde{\pi}(n_i) = \pi_A^*(x_i)$, $y(m_i, \tilde{\pi}(m_i)) = y(m_i, \pi_A^*(x_i))$ is the weight of the branch emanating from m_i for action G in Λ_m . This extends similarly to branches associated with action B and Λ_n . Thus, the three decision trees differ in both the values of the nodes and the weights of the branches. Since π^* is deterministic, so is $\tilde{\pi}$. Therefore, from Equation (13) and Equation (3), x_i , m_i , and n_i are deterministic quantities once the specific path of actions is specified by $\mathcal{G}$.

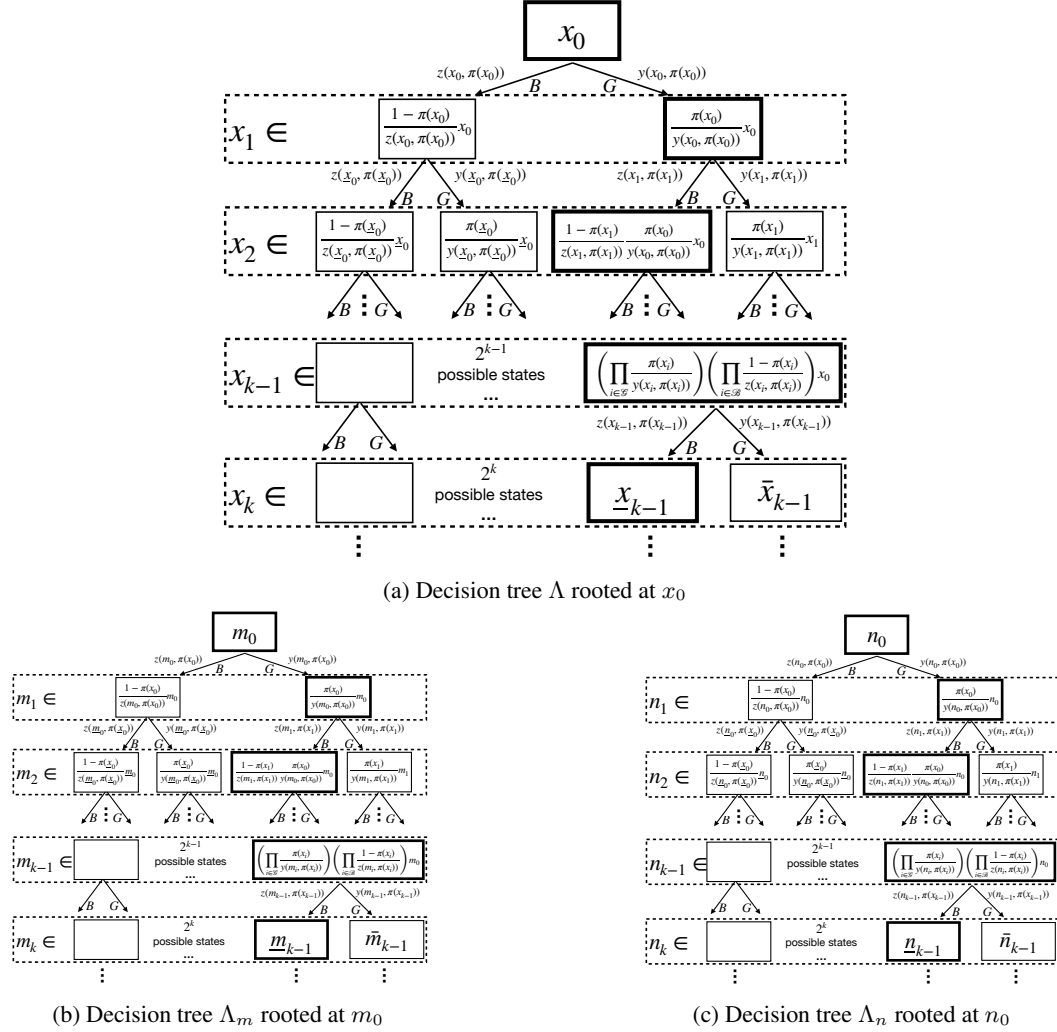


Figure 3: Here we depict the belief evolution in the form of three binary trees. The tree begins with state x_0 , m_0 , and n_0 , respectively, and level i contains 2^i possible states reachable after i epochs. Left branches correspond to action B leading to updated belief $\underline{x}_0$ (respectively, $\underline{m}_0$ and $\underline{n}_0$). Right branches correspond to action G leading to updated belief $\bar{x}_0$ (respectively, $\bar{m}_0$ and $\bar{n}_0$). Each branch is labeled with the corresponding probability, and the bold states indicate an example path, like λ . In the example path, $0 \in \mathcal{G}$ and $1, k-1 \in \mathcal{B}$. The bold states in each level of the three trees are counterparts.

Now let $\phi_k^\pi(b_0) = \mathbb{E}[r_A(b_k, \pi(b_k))]$ be the expected instantaneous reward at epoch k when beginning with state b_0 and applying policy π and $\phi^*(x) = \phi^{\pi^*}(x)$. Applying Fubini's Theorem, we can rewrite our value function

as

$$V_A^\pi(b_0) = \mathbb{E} \left[\sum_{i=0}^{\infty} \delta^i r_A(b_i, \pi(b_i)) \right] = \sum_{i=0}^{\infty} \delta^i \mathbb{E}[r_A(b_i, \pi(b_i))] = \sum_{i=0}^{\infty} \delta^i \phi_i^\pi(b_0) \quad (25)$$

Thus, we can rewrite (24), as

$$\sum_{i=0}^{\infty} \delta^i \phi_i^*(x_0) \leq t \left[\sum_{i=0}^{\infty} \delta^i \phi_i^{\tilde{\pi}}(m_0) \right] + (1-t) \left[\sum_{i=0}^{\infty} \delta^i \phi_i^{\tilde{\pi}}(n_0) \right] \quad (26)$$

To prove this, we show that for all $i = 0, 1, 2, \dots$

$$\phi_i^*(x_0) \leq t \phi_i^{\tilde{\pi}}(m_0) + (1-t) \phi_i^{\tilde{\pi}}(n_0) \quad (27)$$

leading directly to (26), (24), and finally the convexity of $V^*(\cdot)$ with respect to public belief.

Now let us consider general $i = k$. $\phi_{k-1}^\pi(x_0)$ will have 2^{k-1} terms each corresponding to one path of possible actions in the set $\{G, B\}^{k-2}$. Thus, at epoch $k-1$, the path λ will yield the following term in $\phi_{k-1}^*(x_0)$

$$\left(\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(x_i, \pi_A^*(x_i)) \right) r_A(x_{k-1}, \pi_A^*(x_{k-1})) \quad (28)$$

We now define $P_\lambda^*(x_0) = \left(\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(x_i, \pi_A^*(x_i)) \right)$, that is, the probability of following the specified path λ when beginning with state x_0 and following policy π_A^* . Again, we will use $\bar{x}_i$ and $\underline{x}_i$ to represent updated beliefs from x_i after observing action G or B , respectively. Thus, $\bar{x}_i$ and $\underline{x}_i$ constitute belief values of the children nodes (in the $(i+1)$ th level) of the node with belief value x_i in the i -th level. Refer to levels $k-1$ and k in figure 3a. Considering the children $\bar{x}_{k-1}$ and $\underline{x}_{k-1}$ (in the k -th level) of the node x_{k-1} (in the $(k-1)$ -th level), we obtain the following two terms in $\phi_k^*(x_0)$:

$$P_\lambda^*(x_0) \left[y(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \quad (29)$$

Similarly, we define $P_\lambda^{\tilde{\pi}}(m_0) = \left(\prod_{i \in \mathcal{G}} y(m_i, \tilde{\pi}(m_i)) \right) \left(\prod_{i \in \mathcal{B}} z(m_i, \tilde{\pi}(m_i)) \right)$, that is, the probability of following the specified path λ when beginning with state m_0 and following policy $\tilde{\pi}$. Because $\tilde{\pi}(m_i) = \pi_A^*(x_i)$, $P_\lambda^{\tilde{\pi}}(m_0) = \left(\prod_{i \in \mathcal{G}} y(m_i, \pi_A^*(x_i)) \right) \left(\prod_{i \in \mathcal{B}} z(m_i, \pi_A^*(x_i)) \right)$. We also define $P_\lambda^{\tilde{\pi}}(n_0)$ by replacing $\{m_i\}$ with $\{n_i\}$.

The corresponding terms in $\phi_k^{\tilde{\pi}}(m_0)$ and $\phi_k^{\tilde{\pi}}(n_0)$ are

$$P_\lambda^*(m_0) \left[y(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \quad (30)$$

$$P_\lambda^{\tilde{\pi}}(n_0) \left[y(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \quad (31)$$

We will show that

$$\begin{aligned} & P_\lambda^*(x_0) \left[y(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \\ & \leq t P_\lambda^*(m_0) \left[y(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \\ & + (1-t) P_\lambda^{\tilde{\pi}}(n_0) \left[y(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1})) + z(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \right] \end{aligned} \quad (32)$$

Similarly, each pair of children for each term in the $(k-1)$ -th level of the belief tree can be combined and the corresponding inequalities can be obtained following the same process. Summing the resulting inequalities will yield (27), leading to (26), (24), and the convexity of $V^*(\cdot)$.

Proof of (32)

We now focus on the latter of the two terms of (29). We will show the following two properties

$$\begin{aligned} \underline{x}_{k-1} &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{m}_{k-1} \\ &+ (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{n}_{k-1} \end{aligned} \quad (33)$$

$$\begin{aligned}
1 &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\
&+ (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right)
\end{aligned} \tag{34}$$

Defining $\tilde{t} = t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right)$, it follows from (33) and (34) that

$$\underline{x}_{k-1} = \tilde{t} \underline{m}_{k-1} + (1-\tilde{t}) \underline{n}_{k-1} \tag{35}$$

From the expressions for $\tilde{t}$, $P_\lambda^*(x_0)$, $P_\lambda^{\tilde{\pi}}(m_0)$, it follows that:

$$\tilde{t} z(x_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^*(x_0) = t z(m_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^{\tilde{\pi}}(m_0) \tag{36}$$

Similarly, we can show

$$\tilde{t} z(x_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^*(x_0) = (1-t) z(n_{k-1}, \pi_A^*(x_{k-1})) P_\lambda^{\tilde{\pi}}(n_0) \tag{37}$$

We apply this as follows to the latter term of (29):

$$\begin{aligned}
&P_\lambda^*(x_0) z(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{x}_{k-1}, \pi_A^*(\underline{x}_{k-1})) \\
&\leq P_\lambda^*(x_0) z(x_{k-1}, \pi_A^*(x_{k-1})) \\
&\quad [\tilde{t} r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1})) + (1-\tilde{t}) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (\text{Convexity of } r, \text{Eqn 35}) \\
&= t [P_\lambda^{\tilde{\pi}}(m_0) z(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{m}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \\
&\quad + (1-t) [P_\lambda^{\tilde{\pi}}(n_0) z(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\underline{n}_{k-1}, \pi_A^*(\underline{x}_{k-1}))] \quad (\text{Eqns 36, 37})
\end{aligned} \tag{38}$$

Applying the same argument to the first term of (29) yields:

$$\begin{aligned}
&P_\lambda^*(x_0) y(x_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{x}_{k-1}, \pi_A^*(\bar{x}_{k-1})) \\
&\leq t [P_\lambda^{\tilde{\pi}}(m_0) y(m_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{m}_{k-1}, \pi_A^*(\bar{x}_{k-1}))] \\
&\quad + (1-t) [P_\lambda^{\tilde{\pi}}(n_0) y(n_{k-1}, \pi_A^*(x_{k-1})) r_A(\bar{n}_{k-1}, \pi_A^*(\bar{x}_{k-1}))]
\end{aligned} \tag{39}$$

Summing (38) and (39) yields (32).

We now complete the proof of (32) and, therefore, the entire proof, by proving (33) and (34).

Proof of (33)

From the recurrence relation in Equation (13) and Equation (3), we can write $\underline{x}_{k-1}$, $\underline{m}_{k-1}$, and $\underline{n}_{k-1}$ as the product of all belief updates with x_0 , m_0 , and n_0 , respectively, as follows. Each term in the products corresponds to the multiplicative state update after a single action.

$$\underline{x}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(x_i, \pi_A^*(x_i))} \right) x_0 \tag{40}$$

$$\underline{m}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(m_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(m_i, \pi_A^*(x_i))} \right) m_0 \tag{41}$$

$$\underline{n}_{k-1} = \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(n_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(n_i, \pi_A^*(x_i))} \right) n_0 \tag{42}$$

These equations may be best understood via the illustration in figure 3. We then reorganize terms to relate $\underline{x}_{k-1}$ to $\underline{m}_{k-1}$ and $\underline{n}_{k-1}$, using the fact that $x_0 = tm_0 + (1-t)n_0$.

$$\begin{aligned}\underline{x}_{k-1} &= \left(\prod_{i \in \mathcal{G}} \frac{\pi_A^*(x_i)}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{1 - \pi_A^*(x_i)}{z(x_i, \pi_A^*(x_i))} \right) [tm_0 + (1-t)n_0] \quad (\text{from 40}) \\ &= t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{m}_{k-1} \\ &\quad + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \underline{n}_{k-1} \quad (\text{from 41 and 42}) \quad (43)\end{aligned}$$

This proves (33).

Proof of (34)

We first consider the numerator of the first term in (34): $\prod_{i \in \mathcal{G}} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}} z(m_i, \pi_A^*(x_i))$. We now prove that this can be written as $\alpha_0 m_0 + \beta_0$ for some α_0, β_0 which depend only on $\{\pi_A^*(x_i)\}_{i=1, \dots, k-1}$ and the nature of the dependence is determined by the sequence of actions along this path in stages 1 to $k-1$. We will prove this via induction working backwards from the final term (i.e. the $(k-1)$ -th term) of the product. Therefore our base case will be the final term of this product which can be either $y(m_{k-1}, \pi_A^*(x_{k-1}))$ or $z(m_{k-1}, \pi_A^*(x_{k-1}))$, depending on the preceding action (in the $(k-1)$ -th stage).

$$\begin{aligned}y(m_{k-1}, \pi_A^*(x_{k-1})) &= 1 + 2m_{k-1}\pi_A^*(x_{k-1}) - m_{k-1} - \pi_A^*(x_{k-1}) \\ &= (2\pi_A^*(x_{k-1}) - 1)m_{k-1} + (1 - \pi_A^*(x_{k-1})) \\ z(m_{k-1}, \pi_A^*(x_{k-1})) &= m_{k-1} + \pi_A^*(x_{k-1}) - 2m_{k-1}\pi_A^*(x_{k-1}) \\ &= (1 - 2\pi_A^*(x_{k-1}))m_{k-1} + (\pi_A^*(x_{k-1}))\end{aligned}$$

Thus, the final term of the product is of the form $\alpha_{k-1}m_{k-1} + \beta_{k-1}$ where α_{k-1} and β_{k-1} depend only upon $\pi_A^*(x_{k-1})$, and the nature of the dependence is determined entirely by the $(k-1)$ -th action (since this action determines whether the final term is $y(\cdot)$ or $z(\cdot)$).

For the inductive step, assume that $\prod_{i \in \mathcal{G}, i > j} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j} z(m_i, \pi_A^*(x_i))$ can be written as $\alpha_{j+1}m_{j+1} + \beta_{j+1}$ for some $\alpha_{j+1}, \beta_{j+1}$ which depend only on $\{\pi_A^*(x_i)\}_{i > j}$ and the nature of the dependence is determined entirely by the sequence of actions in this path for $i > j$ to $i = k-1$. We will now show that $\prod_{i \in \mathcal{G}, i > j-1} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j-1} z(m_i, \pi_A^*(x_i))$ can be written as $\alpha_j m_j + \beta_j$ again with α_j, β_j depending only on $\{\pi_A^*(x_i)\}_{i > j-1}$ with the nature of the dependence determined by the sequence of actions on

this path for $i > j-1$ to $i = k-1$. We define $f(b, q, a) = \begin{cases} y(b, q) & a = G \\ z(b, q) & a = B \end{cases}$ and $g(q, a) = \begin{cases} q & a = G \\ 1 - q & a = B \end{cases}$.

Thus,

$$\begin{aligned}&\prod_{i \in \mathcal{G}, i > j-1} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j-1} z(m_i, \pi_A^*(x_i)) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) \prod_{i \in \mathcal{G}, i > j} y(m_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}, i > j} z(m_i, \pi_A^*(x_i)) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) [\alpha_{j+1}m_{j+1} + \beta_{j+1}] \quad (\text{inductive hypothesis}) \\ &= f(m_j, \pi_A^*(x_j), a_{j+1}) \left[\alpha_{j+1} \frac{g(\pi_A^*(x_j), a_{j+1})}{f(m_j, \pi_A^*(x_j), a_{j+1})} m_j + \beta_{j+1} \right] \quad (\text{Equation (13)}) \\ &= \alpha_{j+1}g(\pi_A^*(x_j), a_{j+1})m_j + \beta_{j+1}f(m_j, \pi_A^*(x_j), a_{j+1})\end{aligned}$$

Thus, from the definitions of $f(\cdot), g(\cdot), y(\cdot)$, and $z(\cdot)$,

$$\alpha_j = \begin{cases} \alpha_{j+1}\pi_A^*(x_j) + \beta_{j+1}(2\pi_A^*(x_j) - 1) & a_{j+1} = G \\ \alpha_{j+1}(1 - \pi_A^*(x_j)) + \beta_{j+1}(1 - 2\pi_A^*(x_j)) & a_{j+1} = B \end{cases} \text{ and } \beta_j = \begin{cases} \beta_{j+1}(1 - \pi_A^*(x_j)) & a_{j+1} = G \\ \beta_{j+1}\pi_A^*(x_j) & a_{j+1} = B \end{cases}$$

In both cases, by our inductive hypothesis, $\alpha_{j+1}, \beta_{j+1}$ depend only on $\{\pi_A^*(x_i)\}_{i > j}$ and the actions in this path from $i > j$ to $i = k-1$. Thus, the claim holds for α_j, β_j , completing the induction.

Now recall that the denominator of (34) is $\prod_{i \in \mathcal{G}} y(x_i, \pi_A^*(x_i)) \prod_{i \in \mathcal{B} \cup \{k-1\}} z(x_i, \pi_A^*(x_i))$. Note that the policy and sequence of actions is the same as those for the numerator. Thus, because, as shown, the coefficients α_0, β_0 depend only on $\{\pi_A^*(x_i)\}$ and the path of actions, the same logic can be applied to write the denominator

as $\alpha_0 x_0 + \beta_0$ for the same α_0, β_0 . This can be done similarly for the numerator of the second term in (34). Thus, we can reason about the coefficients in (33) as follows:

$$\begin{aligned}
& t \left(\prod_{i \in \mathcal{G}} \frac{y(m_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(m_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\
& + (1-t) \left(\prod_{i \in \mathcal{G}} \frac{y(n_i, \pi_A^*(x_i))}{y(x_i, \pi_A^*(x_i))} \right) \left(\prod_{i \in \mathcal{B} \cup \{k-1\}} \frac{z(n_i, \pi_A^*(x_i))}{z(x_i, \pi_A^*(x_i))} \right) \\
& = \frac{t(\alpha_0 m_0 + \beta_0) + (1-t)(\alpha_0 n_0 + \beta_0)}{\alpha_0 x_0 + \beta_0} \\
& = \frac{\alpha_0 x_0 + \beta_0}{\alpha_0 x_0 + \beta_0} \quad (\text{Since } x_0 = t m_0 + (1-t) n_0) \\
& = 1
\end{aligned}$$

Thus, (34) holds, completing the proof. $\square$

C.4 ALTRUISTIC MYOPIC POLICY BOUND

Lemma 9.

$$\pi_A^0(b) \leq \pi_A^*(b) \quad \forall b \in [0, 1]$$

Assume by way of contradiction that there exists $b \in [0, 1]$ such that $\pi_A^0(b) > \pi_A^*(b)$. We will show that $V_A^*(b) < V_A^\pi(b)$ for a policy π we construct, violating the optimality of $V_A^*(\cdot)$.

Let π be such that, when starting at public belief b , the policy π applies precision $\pi_A^0(b)$ at the current time step and then applies the optimal policy at all future time steps. We now elaborate $V_A^*(b) - V_A^\pi(b)$ by breaking the value function into instantaneous reward r_A and future discounted reward. We will refer to b^+ and b^- as the updated beliefs after receiving signals G and B , respectively when applying the optimal policy and b_m^+ and b_m^- as the same when applying the myopic policy.

$$\begin{aligned}
V_A^*(b) - V_A^\pi(b) &= r_A(b, \pi_A^*(b)) - r_A(b, \pi_A^0(b)) \\
&+ \delta y(b, \pi_A^*(b)) V_A^*(b^+) + \delta z(b, \pi_A^*(b)) V_A^*(b^-) \\
&- \delta y(b, \pi_A^0(b)) V_A^*(b_m^+) - \delta z(b, \pi_A^0(b)) V_A^*(b_m^-) \\
&< \delta y(b, \pi_A^*(b)) V_A^*(b^+) + \delta z(b, \pi_A^*(b)) V_A^*(b^-) \\
&- \delta y(b, \pi_A^0(b)) V_A^*(b_m^+) - \delta z(b, \pi_A^0(b)) V_A^*(b_m^-) \quad (\text{definition of } \pi_A^0) \\
&\leq 0 \quad (\text{Theorem 2})
\end{aligned}$$

Here, we applied the fact that the instantaneous reward under $\pi_A^0(\cdot)$ must be greater from the definition of the myopic optimal policy.

We then relied on the convexity of $V_A^*(\cdot)$ from Theorem 2 for the future cost. From Equation (13), note that $b_m^- \leq b^- \leq b \leq b^+ \leq b_m^+$. Along with this convexity, this leads to the last step above and completes the proof. $\blacksquare$

C.5 PROOF OF THEOREM 3: OPTIMAL ALTRUISTIC POLICY

Consider $V_A^\pi(b)$ with policy π applying precision q in the current time step and applying the optimal policy at all future time steps. The q that maximizes $V_A^\pi(b)$ provides the optimum precision starting at public belief b . Because the planner never benefits from decreasing precision below the baseline p (see Appendix C.9), we restrict q to $[p, 1]$ without loss of generality.

We will let b^+ and b^- denote the positive and negative belief updates possible after receiving a signal with precision q . Taking the second derivative of $V_A^\pi(b)$ with respect to q for $q \geq \max(b, 1-b)$:

$$\frac{\partial^2}{\partial q^2} V_A^\pi(b) = \begin{cases} -\ddot{\beta}(q) + \delta b^2 (1-b)^2 \left[\frac{\ddot{V}_A^*(b^+)}{y^3(b, q)} + \frac{\ddot{V}_A^*(b^-)}{z^3(b, q)} \right] & q \geq \max(b, 1-b) \\ -\ddot{\beta}(q) & \text{otherwise} \end{cases}$$

With the convexity and concavity of $V_A^*(\cdot)$ and $\beta(\cdot)$, respectively, the above shows that $V_A^\pi(b)$ is convex with respect to q on $[\max(b, 1-b), 1]$. Thus, on this interval, it is maximized at one of the two extreme points.

For $q \in [0.5, \max(b, 1 - b))$, the derivative of $V_A^\pi(b)$ with respect to q is negative; thus, the optimal choice will be the baseline precision p to minimize the cost. Thus, there are three possible optimal precisions: p , $\max(b, 1 - b)$, and 1.

Note that the expected value under the latter two candidates is always strictly negative. Under precision p , however, the expected value is 0 at $b = 0$ and $b = 1$. This, combined with the non-positivity and symmetry of the value function (Lemma 8) implies the existence of d_A such that $\pi_A^*(b) = p$ for $b \in [0, d_A] \cup (1 - d_A, 1]$. Furthermore, from Theorem 1 and applying Lemma 9, $d_A \in (0, t_M)$ where t_M is defined as in the statement of Theorem 1.

Finally, Theorem 1 and applying Lemma 9, also imply the existence of $t_A \in [d_A, t_M]$ such that $\pi_A^*(b) = 1$ for $b \in (t_A, 1 - t_A)$. This completes the proof. ■

C.6 PROOF OF THEOREM 4: MYOPIC BIASED POLICY

Preliminaries Applying the action rule from (2), the instantaneous reward (6) can be written as follows:

$$r_B(b, q) = \begin{cases} -\beta(|q - p|) - Cz(b, q) & q \geq \max(b, 1 - b) \\ -\beta(|q - p|) - C & b < 1 - q \\ -\beta(|q - p|) & b > q \end{cases} \quad (44)$$

Note that $z(b, q) = b + q - 2bq$ increases with respect to q when $b < 0.5$ and decreases with respect to q when $b > 0.5$.

At $b = 0.5$, $z(b, q) = z(0.5, q) = 0.5$ regardless of the precision chosen. Furthermore, at this public belief, $q \geq \max(b, 1 - b)$ because $q \geq 0.5$ for any q , falling in the first case of (44). In this case, the second term of $r_B(b, q)$ is unchanging with respect to q , so the planner seeks to maximize only the first term, which is accomplished at $q = p$.

We now prove a series of lemmas (10-14) which will assist us in our proof of Theorem 4.

Lemma 10.

If $\pi_B^0(b) \geq \max(b, 1 - b)$, then $\pi_B^0(b) \in \{\max(b, 1 - b), p, 1\}$.

Proof. The reward is differentiable everywhere except $q = p$. Thus, we can write its derivative with respect to the chosen precision as follows (except at $q = p$):

$$\frac{\partial r_B(b, q)}{\partial q} = \begin{cases} -\text{sign}(q - p)\dot{\beta}(|q - p|) & q \geq \max(b, 1 - b) \\ -C(1 - 2b) & \\ -\text{sign}(q - p)\dot{\beta}(|q - p|) & \text{otherwise} \end{cases} \quad (45)$$

In the former case, we can write the second derivative of the reward with respect to q as follows, for $q \neq p$:

$$\frac{\partial^2}{\partial q^2} r_B(b, q) = -\ddot{\beta}(|q - p|)$$

Because $\beta(\cdot)$ is concave, this expression is non-negative, and $r_B(b, q)$ is convex with respect to q . Thus, the reward is maximized at one of the extreme points $\{\max(b, 1 - b), 1\}$ or p because $r_B(b, q)$ is not differentiable at $q = p$. □

Lemma 11.

For $b \in [0, 1 - p)$, $\pi_B^0(b) \in \{p, 1 - b\}$.

Proof. Because $p > 0.5$, for $b \in [0, 1 - p)$, $\max(b, 1 - b) = 1 - b > p$.

First, consider the case when $\pi_B^0(b) < \max(b, 1 - b) = 1 - b$. From (44), in such instances, the reward is maximized by minimizing $|q - p|$. Thus, the optimal precision lower than $\max(b, 1 - b)$ is p .

Now consider $\pi_B^0(b) \geq \max(b, 1 - b) = 1 - b$. By Lemma 10, $\pi_B^0(b) \in \{1 - b, p, 1\}$. What remains is to rule out $\pi_B^0(b) = 1$. To do so, we compare $r_B(b, 1)$ and $r_B(b, 1 - b)$. Both fall under the first case of (44). Because $\beta(\cdot)$ is increasing in its argument and $p < 1 - b \leq 1$, the first term of the reward from (44) is greater for precision $1 - b$ than for precision 1. Since $z(b, q)$ is increasing with respect to q for $b < 0.5$, $r_B(b, 1) \leq r_B(b, \max(b, 1 - b))$ so we can rule out precision 1. This completes the proof. □

Lemma 12.

For $b \in (p, 1]$, $\pi_B^0(b) = p$.

Proof. The reward is strictly negative for any $q \neq p$ because of the term $-\beta(|q - p|)$. For $b \in (p, 1]$ and $q = p$, $p < \max(b, 1 - b) = b$, leading to the third case of (44) and yielding reward 0. Thus, the optimal precision is p . $\square$

Lemma 13.

For $b \in [1 - p, 0.5]$, $\pi_B^0(b) \in \{p, 1 - b\}$.

Proof. When $p = 0.5$, the interval $[1 - p, 0.5]$ contains only $\{0.5\}$. At $b = 0.5$ $p = 1 - b$, thus $\pi_B^0(b) = p = 1 - b = 0.5$. We deal with $p > 0.5 \geq b$ for the remainder of the proof.

For $b \in [1 - p, 0.5]$, $\max(b, 1 - b) = 1 - b \leq p$. Thus, from the first case of (44), precision p yields reward $r_B(b, p) = -Cz(b, p)$. We now consider two cases: $q < \max(b, 1 - b)$ and $q \geq \max(b, 1 - b)$.

Suppose $q < \max(b, 1 - b) = 1 - b$. This leads to the second case of (44) and yields reward $-\beta(|q - p|) - C$. Furthermore, $z(b, q)$ is increasing with respect to q for $b < 0.5$ and takes values between $z(b, 0.5) = 0.5$ and $z(b, 1) = 1 - b$. Thus, $z(b, q) < 1$ for all q . Therefore, the reward from policy $q < 1 - b$ is strictly lower than $r_B(b, p)$, and we can rule out this case.

When $q \geq \max(b, 1 - b)$, applying Lemma 10 yields that the remaining candidate precisions are $\{1 - b, p, 1\}$. We rule out precision 1 by comparing it to precision p . The first term of the reward from (44) is greater for precision p than for precision 1. Since $z(b, q)$ is increasing with respect to q for $b < 0.5$, the second term is also greater for precision p , and we can rule out precision 1. This completes the proof. $\square$

We now define the notion of ϵ -optimal policies.

Definition 1. ϵ -optimal policies

An ϵ -optimal policy $\pi^\epsilon(b)$ is any policy such that the following holds

$$V^*(b) \leq V^{\pi^\epsilon}(b) + \epsilon$$

for $\epsilon > 0$.

Note that from the definition, any optimal policy is also ϵ -optimal for any $\epsilon > 0$. Thus, $\pi_B^0(b)$ is also ϵ -optimal

Following this definition, let $\pi_B^{\epsilon,0}(b)$ denote an ϵ -optimal myopic policy for the biased planner.

Lemma 14.

For $b \in (0.5, p]$ and any $\epsilon > 0$, there are two possibilities:

1. $\pi_B^0(b) \in \{p, 1\}$, or

2. $\pi_B^{\epsilon,0}(b) = b - \epsilon$

Proof. For $b \in (0.5, p]$, $\max(b, 1 - b) = b \leq p$. We consider two cases: $q \geq \max(b, 1 - b)$ or $q < \max(b, 1 - b)$.

In the first case, Lemma 10 yields that $\pi_B^0(b) \in \{b, p, 1\}$. We rule out precision b by comparing it to precision p . The first terms of the reward from (44) is greater for precision p . Furthermore, since $z(b, q)$ is decreasing with respect to q for $b > 0.5$, the second term of the reward is also greater for precision p . Thus, we can rule out precision b . This yields the first claim.

In the latter case, $q < \max(b, 1 - b) = b$. Therefore, (1) since $q \geq 0.5$, $q \in [0.5, b)$, and (2) the third case of (44) applies leading to reward $-\beta(|q - p|)$. Thus, the reward $r_B(b, q)$ is an increasing function of $q \in [0.5, b)$ because $0.5 < b \leq p$. This leads directly to the second claim. $\square$

Remark 3. As noted in Puterman (1990) Section 2.3.1, an optimal policy may not exist for infinite action spaces because the supremum may not be attained. When an optimal policy does not exist, we instead seek ϵ -optimal policies, i.e., policies that yield reward within $\epsilon > 0$ of the supremum. In particular, this is true for case 2 of Lemma 14.

Proof of Theorem 4 We consider the myopic biased policy in 4 cases: $b \in [0, 1 - p]$, $b \in [1 - p, 0.5]$, $b \in (0.5, p]$, and $b \in (p, 1]$, making use of the fact that $p \geq 0.5$. Based on Lemmas 11-14, these cases have the following candidate policies: $\{p, 1 - b\}$, $\{1 - b, p\}$, $\{1, p, b - \epsilon\}$, and $\{p\}$, respectively.

Case 1: $b \in [0, 1 - p]$, $\pi_B^0(b) \in \{p, 1 - b\}$ (Lemma 11)

We will show the existence of a threshold t_1 such that $\pi_B^0(b) = p$ for $b \in [0, t_1]$ and $\pi_B^0(b) = 1 - b$ for $b \in (t_1, 1 - p)$.

We can express the reward under the two candidate policies as follows by applying (44):

$$\begin{aligned} r_B(b, p) &= -C \\ r_B(b, 1 - b) &= -\beta(1 - b - p) - C(1 - 2b(1 - b)) \end{aligned}$$

The derivative of $r_B(b, 1 - b)$ with respect to b is

$$\frac{\partial}{\partial b} r_B(b, 1 - b) = \dot{\beta}(1 - p - b) + 2C(1 - 2b)$$

which is positive as $\beta(\cdot)$ is increasing and $b < 1 - p \leq 0.5$. Furthermore, when evaluated at $b = 0$ and $b = 1 - p$ we obtain

$$\begin{aligned} r_B(0, 1) &= -\beta(1 - p) - C < r_B(b, p) \\ r_B(1 - p, p) &= -C(1 - 2p(1 - p)) > r_B(b, p) \end{aligned}$$

Thus, $r_B(b, 1 - b)$ is increasing on $[0, 1 - p]$ with $r_B(b, 1 - b) < r_B(b, p)$ for $b = 0$ and vice versa for $b = 1 - p$. Because $\beta(\cdot)$ is continuous, so are both reward functions. Thus, there exists $t_1 \in (0, 1 - p)$ such that $\pi_B^0(b) = p$ for $b \in [0, t_1]$ and $\pi_B^0(b) = 1 - b$ for $b \in (t_1, 1 - p)$.

Case 2: $b \in [1 - p, 0.5]$, $\pi_B^0(b) \in \{1 - b, p\}$ (Lemma 13)

If $p = 0.5$, this interval contains only $\{0.5\}$, and, for $b = 0.5$, $1 - b = p$. Thus, the two candidate policies are equivalent, and the optimal policy is $\pi_B^0(b) = p$. For the remainder of this case, we consider $p > 0.5$.

We now prove the existence of t_2, t_3 such that $\pi_B^0(b) = 1 - b$ for $b \in [t_2, t_3]$, and $\pi_B^0(b) = p$ for $b \in [1 - p, t_2] \cup [t_3, 0.5]$.

We can express the reward under the candidate policies as follows by applying (44):

$$\begin{aligned} r_B(b, 1 - b) &= -\beta(p - 1 + b) - Cz(b, 1 - b) \\ r_B(b, p) &= -Cz(b, p) \end{aligned}$$

This leads to the following condition for the optimal policy to be $1 - b$:

$$\begin{aligned} r_B(b, 1 - b) &\geq r_B(b, p) \\ \beta(p - 1 + b) &\leq C[z(b, p) - z(b, 1 - b)] \\ \beta(p - 1 + b) &\leq C(p - 1 + b)(1 - 2b) \end{aligned} \tag{46}$$

The LHS of (46) is a concave, increasing function of b on $[1 - p, 0.5]$. The RHS of (46) is a concave parabola that increases in the first half of $[1 - p, 0.5]$ and decreases in the second half.

Now consider the two halves of the interval $[1 - p, 0.75 - 0.5p]$ and $[0.75 - 0.5p, 0.5]$. In the first half, the LHS and RHS of (46) are increasing and concave, with the RHS also being a parabola. Thus, in the first half, the two intersect at most two points (one of which is at $b = 1 - p$). In the second half, the LHS of (46) is increasing while the RHS is decreasing. Thus, they intersect at most once. Let the two possible intersections for $b > 1 - p$ be t_2 and t_3 with $t_2 \leq t_3$. Thus, what remains is to determine the optimal on each of $(1 - p, t_2)$, (t_2, t_3) , and $(t_3, 0.5)$. At $b = 0.5$, $r_B(b, p) > r_B(b, 1 - b)$ because $p > 0.5$, thus $\pi_B^0(b) = p$ for $b \in (t_3, 0.5)$. Subsequently, $\pi_B^0(b) = 1 - b$ for (t_2, t_3) , and $\pi_B^0(b) = p$ for $(1 - p, t_2)$.

Case 3: $b \in (0.5, p]$, $\pi_B^{\epsilon, 0}(b) \in \{1, p, b - \epsilon\}$ (Lemma 14)

In this case, we show the existence of $0.5 \leq t_4 \leq t_5 \leq p$ such that the optimal precisions on intervals $(0.5, t_4]$, (t_4, t_5) , and $[t_5, p)$ are p , 1 , and $b - \epsilon$, respectively.

We can express the reward under the candidate policies as follows by applying (44):

$$\begin{aligned} r_B(b, 1) &= -\beta(1 - p) - C(1 - b) \\ r_B(b, p) &= -Cz(b, p) \\ r_B(b, b - \epsilon) &= -\beta(p - b + \epsilon) \end{aligned}$$

When evaluated at $b = p$, $r_B(b, b - \epsilon)$ is arbitrarily close to 0, and, therefore, $b - \epsilon$ is the optimal precision on some interval $(t_5, p]$ for $0.5 \leq t_5 < p$.

Now, note that $r_B(b, 1)$ and $r_B(b, p)$ are linear functions in b while $r_B(b, b - \epsilon)$ is convex in b (from the concavity of β). Thus, $r_B(b, b - \epsilon) - r_B(b, p)$ and $r_B(b, b - \epsilon) - r_B(b, 1)$ are convex and, as a result, have convex level sets. This implies that the set on which both of these expressions are positive, i.e., the set for which $b - \epsilon$ is the optimal policy, must also be convex. Therefore, $b - \epsilon$ cannot be optimal on any subset of $(0.5, t_5)$.

What remains is to compare precisions 1 and p on the interval $(0.5, t_5]$. To do so, we inspect the values of $r_B(b, 1)$ and $r_B(b, p)$ at 0.5 and their derivatives.

$$\begin{aligned} r_B(0.5, 1) &= -\beta(1 - p) - 0.5C \\ r_B(0.5, p) &= -0.5C \end{aligned}$$

Because $r_B(0.5, 1) < r_B(0.5, p)$ there exists interval $(0.5, t_4)$ with $t_4 \leq t_5$ such that $\pi_B^0(b) = p$. Furthermore, because both are linear in b , there is at most one crossover point where their ordering changes. Thus, if $t_4 < t_5$, then $\pi_B^0(b) = 1$ on (t_4, t_5) .

Case 4: $b \in (p, 1]$, $\pi_B^0(b) = p$

This follows directly from Lemma 12 and completes the proof of Theorem 4. ■

C.7 BIASED VALUE FUNCTION MONOTONICITY

Lemma 15. For any $b_1, b_2 \in [0, 1]$ such that $b_1 \leq b_2$,

$$V_B^*(b_1) \leq V_B^*(b_2)$$

Proof. Applying (44), the derivative of the biased reward with respect to b can be written as follows:

$$\frac{\partial}{\partial b} r_B(b, q) = \begin{cases} -C(1 - 2q) & q \geq \max(b, 1 - b) \\ 0 & \text{otherwise} \end{cases} \quad (47)$$

Because $q \geq 0.5$ and $z(b, q) \in [0, 1]$, it follows that $r_B(b, q)$ is increasing with respect to b .

Next, note that the Bayesian nature of belief updates implies that the belief process is a martingale. That is, for any fixed q_i , $\mathbb{E}[b_{i+1}] = b_i$. To demonstrate this consider two cases: (1) $q_i < \max(b_i, 1 - b_i)$, and (2) $q_i \geq \max(b_i, 1 - b_i)$.

In the former case, it follows from (3) that $b_{i+1} = b_i$, satisfying the claim.

In the latter case, (13) gives the distribution of b_{i+1} which yields the same.

The monotonicity of $V_B^*(\cdot)$ then follows from Proposition 5 of Smith & McCardle (2002). □

C.8 PROOF OF THEOREM 5: OPTIMAL BIASED POLICY

We first prove that the optimal value function is continuous in the following lemma:

Lemma 16. Biased value function continuity

$V_B^*(\cdot)$ is continuous w.r.t. b .

Proof. First consider the one-stage value function defined as $V_B^0(b) = \sup_{q \in [0.5, 1]} r_B(b, q)$. From Theorem 4, we know that $\pi_B^{\epsilon, 0}(b) \in \{1 - b, 1, b - \epsilon, p\}$. Thus, we can equivalently write the one-stage value function as follows:

$$\begin{aligned} V_B^0(b) &= \sup_{\epsilon > 0} r_B(b, \pi_B^{\epsilon, 0}(b)) \\ &= \sup_{\epsilon > 0} \max_{q \in \{p, 1 - b, 1, b - \epsilon, p\}} r_B(b, q) \end{aligned} \quad (48)$$

Using the above expression, we now argue that $V_B^0(\cdot)$ is continuous on $b \in [0, 1]$. First, consider $b \leq 0.5$. From Theorem 4, an ϵ -optimal policy applies precision $1 - b$ or p for $b \leq 0.5$. In either case, the instantaneous reward will not depend upon ϵ , and the supremum in (48) can be omitted, leaving $V_B^0(b) = \max_{q \in \{1 - b, p\}} r_B(b, q)$. For $b \leq 0.5$, $r_B(b, 1 - b)$ falls in case one of (44) and is thus continuous with respect to b . On the same interval, $r_B(b, p)$ has one discontinuity at $b = 1 - p$ where it switches from case two to case one of (44). At $b = 1 - p$, Theorem 4 shows that the optimal myopic policy switches from $1 - b$ to p . Both policies fall in the first case of

(44), and, at $b = 1 - p$, $p = 1 - b$, so the controls are equal. Thus, the left and right limits coincide with the optimal value, and $V_B^0(\cdot)$ is continuous.

For $b \geq 0.5$, Theorem 4 gives the candidate ϵ -optimal policies $q \in \{1, b - \epsilon, p\}$. When the inner maximum in (48) is attained for $q = b - \epsilon$, the resulting value is $V_B^0(b) = \sup_{\epsilon > 0} r_B(b, b - \epsilon)$. From the third case of (44), we can rewrite this as $V_B^0(b) = \sup_{\epsilon > 0} -\beta(|b - \epsilon - p|)$. Since $b - \epsilon$ is only selected for $b \leq p$ (Theorem 4), this reduces to $V_B^0(b) = -\beta(|p - b|)$. Thus, exchanging the order of the maximum and supremum in (48), we obtain $V_B^0(b) = \max\{r_B(b, 1), -\beta(|p - b|), r_B(b, p)\}$ for $b \geq 0.5$. For the first expression in this maximum, $r_B(b, 1)$ falls into case one of (44) and is thus continuous. The continuity of $-\beta(|p - b|)$ follows from the continuity of $\beta(\cdot)$. For $r_B(b, p)$, since $b \geq 0.5$, the first case of (44) applies for $b \leq p$ and the third case applies for $b > p$. Thus, $r_B(b, p)$ is potentially discontinuous at $b = p$. Therefore, what remains is to show the continuity of $V_B^0(\cdot)$ at $b = p$.

At $b = p$, the myopic ϵ -optimal policy switches from $q = b - \epsilon$ to $q = p$ by Theorem 4 with $\pi_B^{\epsilon, 0}(p) = b - \epsilon$. Thus, for $b = p$, the inner maximum of (48) is attained at $q = b - \epsilon$ and, from the third case of (44), $V_B^0(p) = \sup_{\epsilon > 0} -\beta(\epsilon) = 0$. Similarly, from Theorem 4, for each $\epsilon > 0$, there exists $t_\epsilon < p$ such that on $[t_\epsilon, p]$ the inner maximum of (48) is attained by $q = b - \epsilon$. On this interval, $V_B^0(b) = \sup_{\epsilon > 0} -\beta(|p - b + \epsilon|) = -\beta(|p - b|)$. Therefore, $\lim_{b \uparrow p} V_B^0(b) = 0$. For $b > p$, the myopic optimal policy is $q = p$ from Theorem 4, and this yields reward $V_B^0(b) = 0$ for all $b > p$ from the third case of (44). This proves the continuity of $V_B^0(\cdot)$ at $b = p$ and completes the proof that $V_B^0(\cdot)$ is continuous on $[0, 1]$.

Applying Theorem 4.2 of Hinderer (2005), then yields the continuity of $V_B^*(\cdot)$. $\square$

Remark 4. Given the continuity of $V_B^*(\cdot)$ with respect to public belief and the fact that the control variable lies in the compact set $[0.5, 1]$, one might conclude that the optimal is guaranteed to exist at all public beliefs. However, there is an important subtlety which prevents such a conclusion: while $V_B^*(\cdot)$ is continuous with respect to public belief, it need not be continuous with respect to precision. Specifically, b_{i+1} is a function of both b_i and q_i (see Equation (3)); thus, $\mathbb{E}[V_B^*(b_{i+1})]$ may be denoted as $\phi(b_i, q_i)$. Now, b_{i+1} need not be continuous with respect to q_i , thus $\phi(\cdot)$ may not be continuous with respect to q_i . Thus, there may be regimes in which the optimal value is not attained by any policy. Indeed, we will show that such cases exist.

Proof of Theorem 5 We now leverage Lemma 16 to prove Theorem 5. We again let $z(b, q) = b + q - 2bq$ be the probability of an agent receiving signal B with precision q and prior public belief b and $y(b, q) = 1 - z(b, q)$ be the corresponding probability of the agent receiving signal G . We abuse notation to allow $V_B(b, q)$ to be the expected utility when beginning at public belief b , applying precision q in the current time step, and then following the optimal policy $\pi_B^*(\cdot)$ in all future steps. We can express $V_B(b, q)$ as follows by applying (13), (3), and (44):

$$V_B(b, q) = \begin{cases} -\beta(|q - p|) - Cz(b, q) \\ + \delta y(b, q) V_B^*(f(b, q, G)) \\ + \delta z(b, q) V_B^*(f(b, q, B)) & q \geq \max(b, 1 - b) \\ -\beta(|q - p|) \\ -C\mathbf{1}(b < 0.5) \\ + \delta V_B^*(b) & \text{otherwise} \end{cases} \quad (49)$$

Consider the latter case of (49) when $q < \max(b, 1 - b)$. Here, the only term affected by the precision policy is $-\beta(|q - p|)$, which is maximized by choosing q as close to p as possible. Thus, if the optimal precision is less than $\max(b, 1 - b)$, then it will be as close to p as possible.

We now consider four cases: (1) $b \in [0, 1 - p]$, (2) $b \in [1 - p, 0.5]$, (3) $b \in [0.5, p]$, and (4) $b \in (p, 1]$.

Case 1: $b \in [0, 1 - p]$, claims (A) and (B)

In this case, we will prove the existence of t_1 such that $\pi_B^*(b) = p$ for $b \leq t_1$. First note that if $\pi^*(b) < \max(b, 1 - b)$, we fall into the second case of Equation (49). Here, the only term affected by the precision policy is $-\beta(|q - p|)$, which is maximized by minimizing $|q - p|$. Because $p \in [0.5, 1]$, when $b \in [0, 1 - p]$, $p < \max(b, 1 - b)$. Thus, for $b \in [0, 1 - p]$, if $\pi_B^*(b) < \max(b, 1 - b)$, then $\pi_B^*(b) = p$.

Now, we evaluate the value function at belief $b = 0$. At $b = 0$, the state of the world is known to be B and, thus, no signal, regardless of its precision, will move the public belief. This can be seen mathematically by substituting $b = 0$ into the updated belief expressions in (13). Furthermore, at $b = 0$, if $q \geq \max(b, 1 - b)$, then $q = 1$. As argued above, if the precision is less than $\max(b, 1 - b)$, the optimal precision is $q = p$. We can compare the resultant value under these two candidate policies as follows from Equation (49):

$$V_B(0, 1) = -\beta(1 - p) - C + \delta V_B^*(0) \quad (50)$$

$$V_B(0, p) = -C + \delta V_B^*(0) \quad (51)$$

Because $V_B(0, 1) < V_B(0, p)$, $\pi_B^*(0) = p$ and $V_B^*(0) = V_B(0, p)$. Thus, from Equation (51), $V_B^*(0) = \frac{-C}{1-\delta}$. By Lemma 15, this is a lower bound for the value function at any public belief. Therefore, and since $V_B^*(b) \leq 0$, $|V_B^*(b)| \leq \frac{C}{1-\delta}$ for all $b \in [0, 1]$.

We now show that there exists an interval $[0, \epsilon)$ with $\epsilon > 0$ on which the optimum of the first case of Equation (49) is less than the optimum of the second case of Equation (49). Thus, in this interval the optimum control lies in the second case of Equation (49) so $\pi_B^*(b) < \max(b, 1-b)$. It follows from our initial argument earlier in this case that $\pi_B^*(b) = p$ in this interval, which would complete the proof.

We define function $g(\cdot)$ to be the maximum of the first case of the RHS of Equation (49):

$$g(b) = \max_{q \geq \max(b, 1-b)} \left[-\beta(|q-p|) - Cz(b, q) + \delta y(b, q) V_B^*(f(b, q, G)) + \delta z(b, q) V_B^*(f(b, q, B)) \right] \quad (52)$$

For $b \in (0, 1)$, $f(b, q, G)$ and $f(b, q, B)$ are both continuous with respect to q from Equation (13). Thus, the maximum in Equation (52) must exist because it is of a function which is continuous with respect to q over a compact set $q \in [\max(b, 1-b), 1]$.

We now take the limit of $g(\epsilon)$ as ϵ tends to 0. Because $q \in [\max(\epsilon, 1-\epsilon), 1]$, when $\epsilon \rightarrow 0$, $q \rightarrow 1$ (inside the maximum in Equation (52)). We now reason about each of the four terms in Equation (52). As $\epsilon \rightarrow 0$, $q \rightarrow 1$ so that 1) from the continuity of $\beta(\cdot)$, $\beta(|q-p|) \rightarrow \beta(1-p)$, 2) from the continuity of $z(\cdot)$, $z(\epsilon, q) \rightarrow 1$, $y(\epsilon, q) \rightarrow 0$. Thus, the second term goes to C . Because $V_B^*(\cdot)$ is bounded and $y(\epsilon, q) \rightarrow 0$, the third term goes to 0. Finally, because $z(\epsilon, q) \rightarrow 1$ and $f(\epsilon, q, B) \rightarrow 0$ and since $V_B^*(\cdot)$ is continuous from Lemma 16, the last term approaches $\delta V_B^*(0)$. Thus,

$$\lim_{\epsilon \rightarrow 0} g(\epsilon) = -\beta(1-p) - C + \delta V_B^*(0) \quad (53)$$

Applying precision p , however, yields the following from the second case of Equation (49):

$$\lim_{\epsilon \rightarrow 0} V_B(\epsilon, p) = \lim_{\epsilon \rightarrow 0} [-\beta(p-p) - C + \delta V_B^*(\epsilon)] \quad (54)$$

$$= -C + \delta V_B^*(0) \quad (55)$$

The last equality follows from the continuity of $V_B^*(\cdot)$ from Lemma 16.

Note that the expression in Equation (53) is strictly less than that in Equation (55). Thus, there exists an interval $[0, \epsilon)$ with $\epsilon > 0$ on which the optimum of the first case of Equation (49) is less than the optimum of the second case of Equation (49). The result follows.

Case 2: $b \in [1-p, 0.5)$, Claim (C)

We now argue that for $b_i \in [1-p, 0.5)$, $\pi^*(b_i) \geq 1-b_i$, yielding claim (C). Consider a stationary policy $\pi(\cdot)$ such that $\pi(b_i) < 1-b_i$. Note that, from Equation (3), the public belief will remain unchanged under such a policy. Thus, the same control will be applied at every future time step, and, from Equation (2), every agent will select action B . Using the facts that $1-b_i \leq p$ in this belief range, $\mathbb{P}(a_i = B | b_i, \pi(b_i)) = 1$, and $b_{i+1} = b_i$, we can apply Equation (6) to write the expected total discounted utility under such a policy as follows:

$$V_B^\pi(b_i) = -\beta(p - \pi(b_i)) - C + \delta V_B^\pi(b_i)$$

Thus, if this policy is applied, it will result in expected value $V_B^\pi(b_i) = \frac{-\beta(p - \pi(b_i)) - C}{1-\delta}$. That is, the cost of every agent choosing action B and continuing to apply precision $\pi(b_i)$. This is strictly lower than the value function lower bound of $\frac{-C}{1-\delta}$ established using Lemma 15 in Case 1 because $\pi(b_i) < 1-b_i \leq p$. Therefore, if an optimal policy exists at this point, it must be the case that $\pi_B^*(b_i) \geq 1-b_i$.

The remaining candidate policies are $q_i \in [1-b_i, 1]$. Applying Bellman's principle we can write the following:

$$V_B^*(b_i) = \sup_{q_i \in [1-b_i, 1]} r_B(b_i, q_i) + \delta \mathbb{E}[V_B^*(b_{i+1})] \quad (56)$$

Because $q_i \geq \max(b_i, 1-b_i)$ and $b \in [1-p, 0.5) \subset (0, 1)$, Equation (13) and Equation (3) imply that $b_{i+1} = \frac{b_i q_i}{y(b_i, q_i)}$ or $b_{i+1} = \frac{b_i(1-q_i)}{z(b_i, q_i)}$. Both of these expressions are continuous with respect to q_i and $V_B^*(\cdot)$ is continuous by Lemma 16, thus the RHS of Equation (56) inside the supremum is continuous with respect to q_i . Therefore, in Equation (56), the supremum over a compact set $q_i \in [1-b_i, 1]$ of a function which is continuous with respect to q_i must be attained, implying the existence of an optimal policy. Thus, $\pi_B^*(b_i) \geq 1-b_i$.

Case 3: $b \in [0.5, p]$ Claims (D) and (E)

For $b = 0.5$, $q \geq b$ as we assumed $q \in [0.5, 1]$. This places us in the first case of Equation (49). Furthermore, $\beta(\cdot)$, $y(b, q)$, $z(b, q)$, $f(b, q, G)$, and $f(b, q, B)$ are all continuous with respect to q for $b = 0.5$. Thus, the

supremum over $q \in [0.5, 1]$ of $V(b, q)$ must be attained as it is of a continuous function of q over a compact set of q . Therefore, an optimal policy must exist at $b = 0.5$.

Now consider $b \in (0.5, p]$ and a stationary policy $\pi'(\cdot)$ such that $\pi'(b) < b$. From Equation (3) and Equation (2), applying such a policy will result in public belief remaining unchanged, every agent selecting action G , and the same control being applied at every future time step. Using the facts that $b_i \leq p$, $\mathbb{P}(a_i = G | b_i, \pi'(b_i)) = 1$, and $b_{i+1} = b_i$, we can apply Equation (6) to write the total expected discounted utility under such a policy as follows:

$$V_B^{\pi'}(b_i) = -\beta(p - \pi'(b_i)) + \delta V_B^{\pi'}(b_i)$$

Thus, the policy results in expected value $V_B^{\pi'}(b_i) = \frac{-\beta(p - \pi'(b_i))}{1 - \delta}$. Note that, in this case, $V_B^{\pi'}(b_i)$ is an increasing function of $\pi'(b_i)$. Thus, the supremum of $V^*(\pi')_B(b_i)$ over $\pi'(b_i) \in (0.5, b)$ is not attained.

We now consider two cases: (1) $\max_{q_i \in [b_i, 1]} V(b_i, q_i) \geq \sup_{\pi'(b_i) \in (0.5, b_i)} \frac{-\beta(p - \pi'(b_i))}{1 - \delta}$, and (2) the complement. In the former case, we can argue as we did for $b = 0.5$ that the supremum must be attained and an optimal policy exists such that $\pi_B^*(b_i) \geq b_i$. In the latter case, as argued above, no optimal policy exists. Instead, for sufficiently small $\epsilon > 0$, there exists an ϵ -optimal policy such that $\pi_B^\epsilon(b_i) = b_i - \epsilon$. This yields claims (D) and (E).

What remains is to show that there exists $t_2 \in (0.5, p)$ such that for $b_i \in (t_2, p]$ we fall into the latter case, yielding claim (F). We define functions $m(\cdot)$ and $n(\cdot)$ as follows corresponding to the resultant value from choosing precisions less than b or greater than or equal to b , respectively:

$$\begin{aligned} m(b) &= \sup_{\pi'(b) \in [0.5, b)} \frac{-\beta(p - \pi'(b))}{1 - \delta} \\ n(b) &= \max_{q \geq b} V(b, q) \end{aligned}$$

Taking the limit of $m(\cdot)$ as $b \rightarrow p$ yields:

$$\lim_{b \rightarrow p} m(b) = 0 \tag{57}$$

When $q \geq b$ we fall into the first case of Equation (49). Thus we can expand $n(b)$ as follows:

$$\begin{aligned} n(b) &= \max_{q \geq b} [-\beta(|q - p|) - Cz(b, q) \\ &\quad + \delta y(b, q)V_B^*(b, q) \\ &\quad + \delta z(b, q)V_B^*(b, q)] \end{aligned} \tag{58}$$

We will argue that, for $b \in (0.5, p]$ and $q \geq b$, $n(b) \leq -C(1 - p)$. Since p is a fixed parameter strictly less than 1, this means from Equation (57) that $n(b) \leq -C(1 - p) < \lim_{b \rightarrow p} m(b)$, which implies the existence of an open interval $(t_2, p]$ where an optimal policy does not exist, yielding claim (F). We now argue that, for $b \in (0.5, p]$ and $q \geq b$, $n(b) \leq -C(1 - p)$. Note that every term of Equation (58) is bounded above by 0, so it suffices to show that any one term, say the second term $-Cz(b, q)$, is upper bounded by $-C(1 - p)$. First note that $z(b, q) = q(1 - 2b) + b$ so $z(b, q)$ is decreasing in q since $b > 0.5$. Thus, $z(b, q) \geq z(b, 1) = 1 - b \geq 1 - p$. The last inequality follows because $b \in (0.5, p]$. The claim follows since $p < 1$.

Along with Equation (57), this implies the existence of an open interval (t_2, p) where an optimal policy does not exist, yielding claim (F).

Case 4: $b \in (p, 1]$ Claim (A)

For $b \in (p, 1]$, consider applying stationary policy $\pi(\cdot)$ such that $\pi(b) = p$. Because $b > p$, Equation (3) and Equation (2) imply that such a policy results in public belief remaining unchanged, every agent selecting action G , and the same control being applied at every future time step. We can apply Equation (6) to write

$$V_B^\pi(b) = -\beta(p - p) + \delta V_B^\pi(b) = \delta V_B^\pi(b)$$

Thus, $V_B^\pi(b) = 0$. Since this is also an upper bound of the optimal value function for any b , $\pi_B^*(b) = p$. ■

C.9 SOCIAL WELFARE MONOTONICITY

Lemma 17. *Precision Monotonicity of Social Welfare*

If $\pi_1(b) \leq \pi_2(b) \forall b \in [0, 1]$, for policies $\pi_1(\cdot), \pi_2(\cdot)$, then

$$W^{\pi_1}(b) \leq W^{\pi_2}(b) \forall b \in [0, 1]$$

Proof. It is clear that the one-step welfare $-C \min(b_i, 1 - b_i, \pi(b_i))$ is increasing in $\pi(b_i)$.

Furthermore, note that the convexity of social welfare with respect to public belief follows from the same argument as in the proof of Theorem 2. This is because social welfare is equivalent to the altruistic value function in the special case where $\beta(\cdot) = 0$.

We can then follow the same argument as in the proof of Lemma 9 (Appendix C.4) to show that increasing precision increases welfare. $\square$

D EXTENSION TO HETEROGENEOUS AGENT PREFERENCES

This appendix extension shows that the altruistic-planner contribution is robust to a simple form of heterogeneous preferences: two observable agent types with opposed preferred actions.

Assume two agent types, G and B , with agent i 's type denoted t_i . Type G receives unit reward when $a_i = \omega$ and no reward otherwise; type B receives unit reward when $a_i \neq \omega$ and no reward otherwise. All agents know their own type and all other agents' types, and all other model components remain unchanged.

The private belief update is unchanged because it depends only on the signal structure, $\mathbb{P}(s_i = \omega) = q_i$, not on the agent's payoff type:

$$\tilde{b}_i = \mathbb{P}(\omega = G | \mathcal{H}_i, q_i, s_i) = \begin{cases} \frac{q}{1+2b_iq_i-b_i-q_i} b_i & s_i = G \\ \frac{1-q}{b_i+q_i-2b_iq_i} b_i & s_i = B \end{cases} \quad (59)$$

Using Equation (59), the type-specific action rule is:

$$a_i^G = \begin{cases} s_i & 1 - q_i \leq b_i \leq q_i \\ G & q_i < b_i \\ B & b_i < 1 - q_i \end{cases}$$

$$a_i^B = \begin{cases} !s_i & 1 - q_i \leq b_i \leq q_i \\ B & q_i < b_i \\ G & b_i < 1 - q_i \end{cases}$$

Because each agent's type is publicly known, $q_i \geq \max(b_i, 1 - b_i)$ allows all observers to infer agent i 's private signal perfectly. If $q_i < \max(b_i, 1 - b_i)$, the action is uninformative.

The public-belief update after observing type t_i , precision q_i , and action a_i is therefore:

$$b_{i+1} = f(b_i, q_i) = \begin{cases} \tilde{b}_i & 1 - q_i \leq b_i \leq q_i \\ b_i & \text{o.w.} \end{cases} \quad (60)$$

For heterogeneous but observable types, the altruistic planner's objective remains agent-specific correctness: induce type- G agents to choose ω and type- B agents to choose $!\omega$, matching each type's utility function.

The corresponding instantaneous rewards, with superscripts denoting agent type, are:

$$r^G(b_i, q_i) = -\beta(|q_i - p|) - C \mathbb{P}(a_i \neq \omega)$$

$$= \begin{cases} -\beta(|q_i - p|) - C(1 - q_i) & q_i \geq \max(b_i, 1 - b_i) \\ -\beta(|q_i - p|) - C \min(b_i, 1 - b_i) & q_i < \max(b_i, 1 - b_i) \end{cases}$$

$$r^B(b_i, q_i) = -\beta(|q_i - p|) - C \mathbb{P}(a_i = \omega)$$

$$= \begin{cases} -\beta(|q_i - p|) - C(1 - q_i) & q_i \geq \max(b_i, 1 - b_i) \\ -\beta(|q_i - p|) - C \min(b_i, 1 - b_i) & q_i < \max(b_i, 1 - b_i) \end{cases}$$

The two expressions are identical and equal to Equation (5), so the altruistic planner's problem is equivalent under heterogeneous but observable types. $\blacksquare$

E EXPERIMENTS

This appendix section records the experimental details behind the LLM-based contribution in the main text: the simulation environment, parameter grid, prompts, and additional validation checks used to compare LLM planner behavior with the analytical controlled-learning model.

E.1 EXPERIMENTAL SETUP

The appendix experiment description mirrors the paper’s contribution framing. We simulate controlled social learning with agents deciding whether to buy a new car after observing past actions and an ad whose precision is selected by the planner. LLMs play three roles: agents, planner, and oracle. Each agent receives a profile with demographic information, car use cases, and preferences. At the start of an instance, the offered car is selected to be good or bad. The planner and agents have the information sets and objectives specified in Section 3.

The oracle supports the precision-control contribution in two ways: it generates the generic baseline message used without planner intervention, and, after the planner chooses a precision, it generates a target-agent-specific message with that prescribed precision.

The oracle receives a car fact sheet (safety rating, customer satisfaction, etc.) for each car type and the target agent’s profile. These inputs let it generate a tailored message at the planner’s requested precision. Appendix E.2 gives the setup and prompts.

Each agent makes a binary buy/not-buy decision from the observed decision history and private signal, with the goal of matching the true car-quality state.

The planner observes the same decision history and chooses each agent’s message precision. It also knows the precision of a generic untailored message that can be sent at no cost. The altruistic planner seeks correct decisions, while the biased planner seeks buy decisions.

We compare LLM simulation outcomes against analytical solutions computed with JuliaPOMDP (Egorov et al., 2017).

E.2 PARAMETERS AND PROMPTS

E.2.1 INSTANCE PARAMETERS

All LLM experiments use the LG AI EXAONE Deep 32B model (LG AI Research, 2025). We solve the analytical model and run LLM simulations over the parameter grid in Table 1.

Parameter	Description	Values
C	Cost of a bad action	1
k	Scaling factor for control cost	{0.1, 0.3, 0.5, 0.7, 0.9}
p	Baseline signal precision	{0.6, 0.7, 0.8, 0.9}
δ	Discount Factor	{0, 0.25, 0.5, 0.75, 1}

Table 1: Parameter grid used for both analytical solutions and LLM simulations.

The car fact sheets ensure that the good car is preferable for every profile and the bad car is undesirable for every profile. The GOOD and BAD fact sheets are listed below.

```
fact_sheet["GOOD"] = [
    "EPA-rated range: 290 miles.",
    "Fast-charge 10-80 %% in 24 min on 150 kW DC.",
    "8-year / 100,000-mile battery warranty.",
    "NHTSA 5-Star overall safety rating.",
    "Cold-soak -5 °F keeps 93 %% range.",
    "Cools cabin 100→72 °F in 3 min.",
    "Cargo volume 26 ft³ total.",
    "Tow's 1 500 lb with factory hitch.",
    "Average 1.2 service visits first 3 yrs.",
    "92 %% owners would buy again (Apr-25).",
    "Qualifies for federal credit.",
    "Quarterly OTA updates incl. battery pre-heat."
]
fact_sheet["BAD"] = [
    "EPA-rated range: 155 mi (combined).",
    "Fast-charge 10 %% → 80 %% in 51 min (50 kW DC hardware limit).",
    "Battery warranty: 3 yr / 36 000 mi, first owner only.",
    "NHTSA overall safety rating 3 Stars (frontal-offset weakness).",
    "Cold-soak 15 °F retains 58 %% range; cabin heat cuts range 20%.",
    "Cabin cools 105 °F → 72 °F in 11 min.",
```

```

1836         "Cargo volume 17 ft3; rear seats fixed (no fold-flat).",
1837         "No factory towing approval.",
1838         "Average 3.8 service visits per year (battery-alert issues).",
1839         'Owner survey: 41 %% "would buy again" (1 200 owners, Feb-25).',
1840         "Not eligible for federal tax credit.",
1841         "OTA updates limited to infotainment."
1842     ]

```

Agent profiles were generated programmatically and include name, age, city of residence, interests, planned car use case (e.g., commuting or road trips), and major concerns (e.g., fuel efficiency or winter driving).

E.2.2 PROMPTS

Agent LLM Prompt:

```

1849 sys = (
1850     f"You are deciding whether to buy a car. "
1851     f"Your profile is {profile}"
1852     f"You start believing there's a 50% chance the car is GOOD. "
1853     f"You have two information sources: "
1854     f"1. Previous agents' decisions "
1855     f"2. An advisor's message about the car "
1856     f>Your goal is to maximize your reward "
1857     f"i.e. BUY if the car is GOOD and "
1858     f"NOT BUY if the car is BAD. "
1859     f"If you BUY a GOOD car, you GAIN ${C}. "
1860     f"If you NOT BUY a BAD car, you GAIN ${C}. "
1861     f"If you BUY a BAD car, you LOSE ${C}. "
1862     f"If you NOT BUY a GOOD car, you LOSE ${C}. "
1863     f"belief = your probability (0.0-1.0) that the car is GOOD. "
1864     f"decision = exactly 'BUY' or 'NOT BUY'."
1865 )
1866 user = (
1867     f"PREVIOUS DECISIONS:\n{history_txt or '(none)'}\n\n"
1868     f"ADVISOR MESSAGE:\n{message}\n\n"
1869     f"TASK: Use the previous decisions and the advisor message. "
1870     f"Decide BUY or NOT BUY. "
1871     f"Update your belief about car quality (0.0-1.0). \n\n"
1872     f'JSON response:{{"belief": <number>,"decision": "<BUY or NOT BUY>"}}'
1873 )

```

Planner LLM Prompt:

```

1872 sys = (
1873     f"You are providing messages to potential car buyers. "
1874     f"Each potential car buyer (or agent) has the same information. "
1875     f>Your goal is to get agents to {objective}."
1876     f"Each decision according to your objective, earns you {C} dollars. "
1877     f>You and the agents have the same information. "
1878     f"Initially there is a 50% chance the car is GOOD. "
1879     f"The agents' aim to BUY if the car is GOOD "
1880     f"and NOT BUY if the car is BAD. "
1881     f>You can send a baseline message for free "
1882     f"or tailor the message to the agent at a cost. "
1883     f"Tailoring costs {k} dollars times the change in precision. "
1884     f"(Message precision is the probability (0.5 to 1.0) "
1885     f"that the message leaves the buyer with the correct impression. "
1886     f"For example, if the car is likely GOOD: "
1887     f"a high precision message will likely suggest BUY "
1888     f"and a low precision message will likely suggest NOT BUY. "
1889     f>You may choose to increase or decrease precision. "
1890     f"Agents decide based on your message and the decision history. "
1891     f"They update beliefs by observing previous decisions. "
1892     f"IMPORTANT: Thus your objective is to maximize:"
1893     f"{C} * (# of agents who choose according to your objective) "

```

```

1890         f"- sum of ({d}^i*{k}*abs(message_precision - baseline_precision))\n"
1891         f"where {d} is your discount factor for future cost. "
1892     )
1893     user = (
1894         f"BASELINE MESSAGE precision:\n{pi_base:.2f}\n"
1895         f"DECISION HISTORY:\n{history_txt}\n"
1896         f"TASK:"
1897         f"Based on the history and your objective, "
1898         f"choose your desired precision.\n"
1899         f'JSON response: {{"message_precision": <number>}}\n'
1900     )

```

Oracle LLM Prompt:

```

1903 sys = (
1904     f"You will be given "
1905     f"1. an agent profile, "
1906     f"2. information about a car, "
1907     f"3. the true quality of the car, "
1908     f"4. a baseline message and the 'precision' of that message, and "
1909     f"5. a desired precision for the message you output. "
1910     f"(Message precision is the probability (0.5 to 1.0) "
1911     f"that the message leaves the buyer with the correct impression. "
1912     f"For example, if the car is likely GOOD: "
1913     f"a high precision message will likely suggest BUY "
1914     f"and a low precision message will likely suggest NOT BUY. "
1915     f"IMPORTANT: Output a message with the requested precision. "
1916     f"IMPORTANT: The message should be tailored to the profile. "
1917 )
1918 user = (
1919     f"AGENT PROFILE:{profile}\n"
1920     f"TRUE CAR QUALITY: {omega}\n\n"
1921     f"CAR FACTS: {car_facts}\n"
1922     f"BASELINE MESSAGE:{baseline_message}}\n"
1923     f"BASELINE MESSAGE precision: {p}\n"
1924     f"DESIRED MESSAGE precision: {q}\n"
1925     f"IMPORTANT: Output a message with the requested precision. "
1926     f"TASK: Write a message (MAX 100 words) with the desired precision."
1927     f'JSON response: {{"message": "<message>"}}'
1928 )

```

E.3 VALIDATION OF LLM BELIEFS AND ORACLE

We add two validation checks for the empirical contribution: the self-reported beliefs used in figure 1b and the oracle’s ability to generate messages at requested precision. These checks treat LLMs as rational and utility-maximizing, but not necessarily Bayesian. The first verifies that a reported posterior is the belief governing state-contingent choices; the second verifies that oracle-generated messages achieve the intended precision.

First, we give the LLM a prior and a new signal, elicit its posterior, and then ask for a binary “buy/not-buy” decision that yields unit reward if and only if the state is good. A rational agent buys exactly when its subjective belief exceeds 0.5. As shown in figure 4a, the LLM’s choices align sharply with this threshold: it overwhelmingly buys above 0.5 and does not buy below it. Second, we elicit willingness to pay for a payoff contingent on the true state. For prices c , the agent decides whether to pay c in exchange for unit utility if the state is good. A rational agent with belief p accepts exactly when $c < p$. Figure 4b plots buy rates across posterior–price pairs and shows high acceptance when $c < p$ and low acceptance when $c > p$, as expected for a utility-maximizing decision maker with belief p . Together, the two checks support using self-reported posterior beliefs as the model’s decision-relevant beliefs.

We also validate oracle precision. After the oracle generates a message with requested precision q , another LLM estimates the realized precision $\hat{q}$. We evaluate $q \in \{0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}$ across 100 programmatically generated profiles and 10 trials per combination, for 6,000 trials total. The mean absolute error $|q - \hat{q}|$ is 0.025, indicating that realized message precision closely follows requested precision.

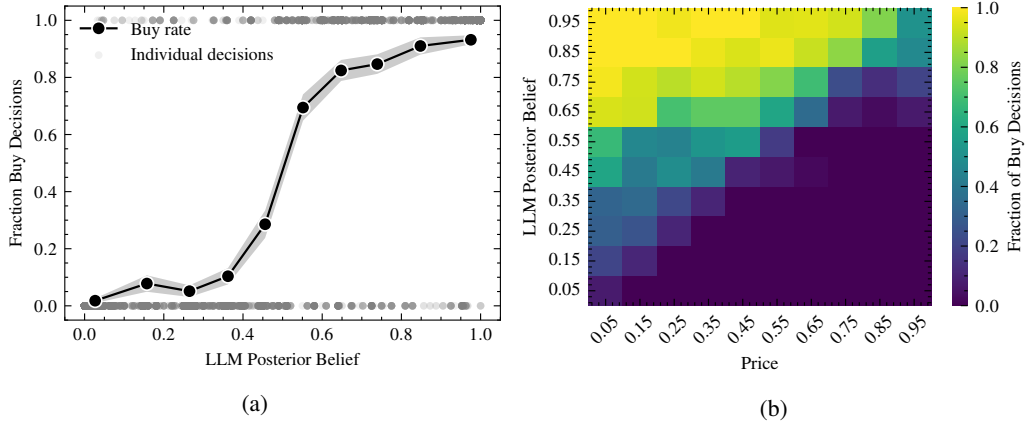


Figure 4: Validation that LLM self-reported beliefs govern choices: buy rates as a function of (a) self-reported posterior belief and (b) self-reported posterior plus buy price.

E.4 IMPACT OF NON-BAYESIAN LLM BELIEF UPDATES ON LEARNING TRAJECTORIES

We finally use belief trajectories from one LLM experiment to show how non-Bayesian updating affects the social-learning path. We fix the policy to the analytically optimal policy and apply it to LLM-simulated agents. When the car is bad (right panel), the altruistic planner drives the Bayesian belief (dotted black line) rapidly toward 0, while the LLM belief (solid black line) remains more stubborn. The aggregate trajectory follows from the micro-level biases: public belief resists entering a cascade (NB3) because individual agents overreact to occasional positive signals that contradict the group’s increasingly negative belief (NB2).

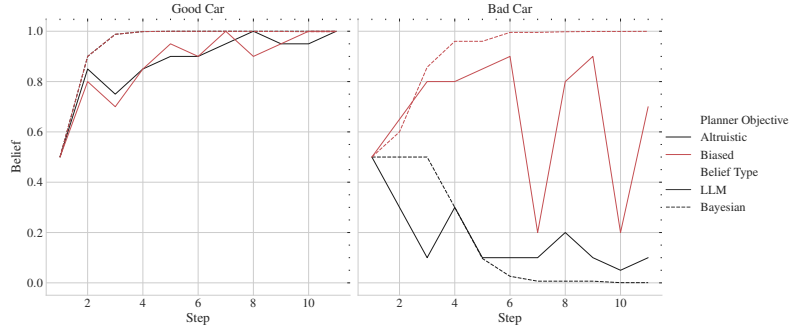


Figure 5: Belief trajectories comparing true Bayesian belief with LLM-reported belief under analytically optimal policies across planner objectives and true states.